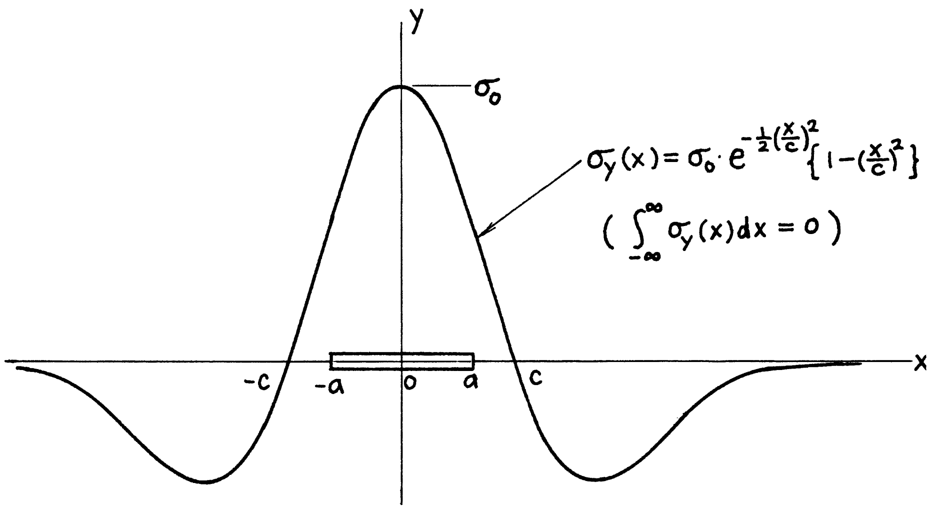


$$\alpha = \frac{a}{c}$$

$$K_I = \sigma_0 \sqrt{\pi a}\, e^{-.42\alpha^2} \left(1 - \frac{1}{\pi}\alpha^2\right)$$

Crack Opening Area:

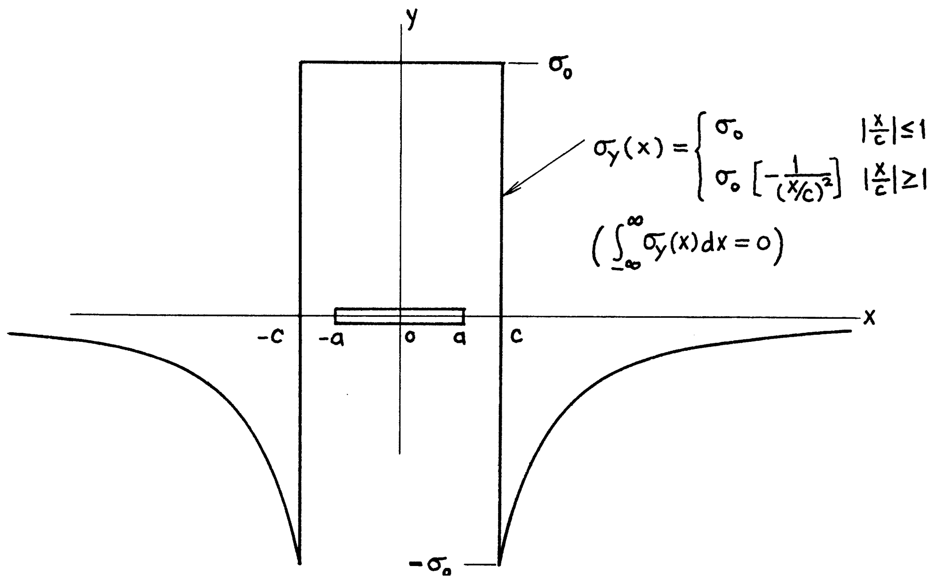
$$A = \frac{2\sigma_0 \pi a^2}{E'} e^{-.36\alpha^2}$$

Opening at Center: $\delta = 2v(0, 0)$

$$\delta = \frac{4\sigma_0 a}{E'} \cdot \frac{1}{1 + \frac{1}{4}\alpha^2}$$

Method: K_I Integration of **page 5.11**
A and δ Paris' Equation (see **Appendix B**)
Accuracy: K_I 0.5 %
A 1 % for $a/c \le \sqrt{\pi}$
δ 2 % for $a/c \le \sqrt{\pi}$
References: **Terada 1976, Tada 1983b, 1985**

NOTE: $K_I < 0$ for $a/c \gtrsim \sqrt{\pi}$



$$\alpha = \frac{a}{c}$$

$$K_I = \sigma_0\sqrt{\pi a}\begin{cases} 1 & \alpha \le 1 \\ \frac{2}{\pi}\left(\sin^{-1}\frac{1}{\alpha} - \frac{\sqrt{\alpha^2-1}}{\alpha^2}\right) & \alpha \ge 1 \end{cases}$$

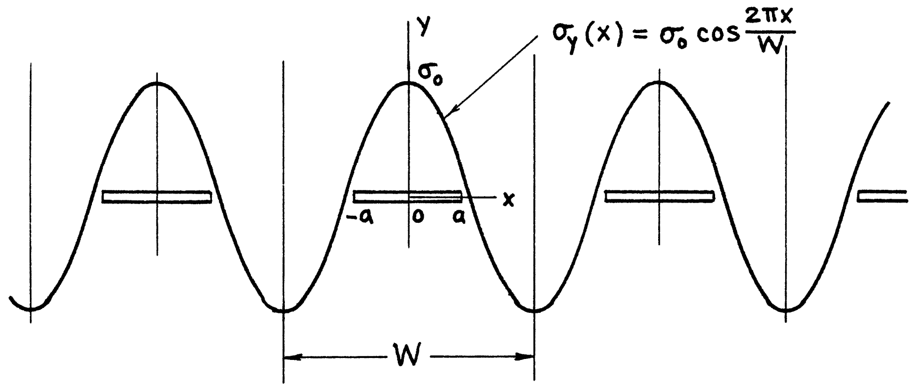
Crack Opening Area:

$$A = \frac{2\,\sigma_0\,\pi a^2}{E'}\begin{cases} 1 & \alpha \le 1 \\ \frac{2}{\pi}\left(\frac{2}{\alpha^2}\cos^{-1}\frac{1}{\alpha} + \sin^{-1}\frac{1}{\alpha} - \frac{\sqrt{\alpha^2-1}}{\alpha^2}\right) & \alpha \ge 1 \end{cases}$$

Opening at Center: $\delta = 2v(0,0)$

$$\delta = \frac{4\,\sigma_0\,a}{E'}\begin{cases} 1 & \alpha \le 1 \\ \frac{2}{\pi}\left(\sin^{-1}\frac{1}{\alpha} + \frac{\sqrt{\alpha^2-1}}{\alpha^2}\right) & \alpha \ge 1 \end{cases}$$

Method: K_I Integration of **page 5.11**
A and δ Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

$$Z_I(z) = \sigma_0 \frac{\sin \frac{\pi z}{W}}{\sqrt{\left(\sin \frac{\pi z}{W}\right)^2 - \left(\sin \frac{\pi a}{W}\right)^2}} \left[2\left(\cos \frac{\pi z}{W}\right)^2 - \left(\cos \frac{\pi a}{W}\right)^2\right]$$

$$\overline{Z}_I(z) = \sigma_0 \frac{W}{\pi} \cos \frac{\pi z}{W} \sqrt{\left(\sin \frac{\pi z}{W}\right)^2 - \left(\sin \frac{\pi a}{W}\right)^2}$$

$$K_I = \sigma_0 \sqrt{\pi a} \sqrt{\frac{W}{\pi a} \tan \frac{\pi a}{W}} \left(\cos \frac{\pi a}{W}\right)^2$$

$$\underset{|x| \le a}{\mathrm{Im}} \left\{\overline{Z}(x)\right\} = \frac{\sigma_0 W}{\pi} \cos \frac{\pi x}{W} \sqrt{\left(\sin \frac{\pi a}{W}\right)^2 - \left(\sin \frac{\pi x}{W}\right)^2}$$

Crack Opening Area:

$$A = \frac{2\sigma_0 W^2}{\pi E'} \left(\sin \frac{\pi a}{W}\right)^2$$

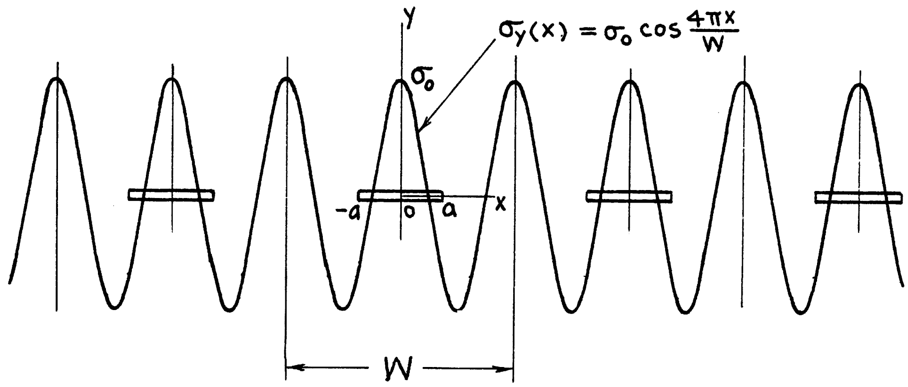
Opening at Center: $\delta = 2v(0,\ 0)$

$$\delta = \frac{4\sigma_0 W}{\pi E'} \sin \frac{\pi a}{W}$$

Relative Vertical Displacement at Infinity due to Crack:

$$\Delta\,(= \Delta_{crack}) = {}^{A}\!/_{W} \left(= \frac{1}{2} \delta \sin \frac{\pi a}{W}\right)$$

Methods: Westergaard Stress Function (Integration of **page 7.7**)
A, δ, Δ also Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**



$z = x + iy$

$$Z_I(z) = \sigma_0 \frac{\sin\frac{\pi z}{W}}{\sqrt{\left(\sin\frac{\pi z}{W}\right)^2 - \left(\sin\frac{\pi a}{W}\right)^2}} \left[8\left(\cos\frac{\pi z}{W}\right)^4 - 4\left(\cos\frac{\pi z}{W}\right)^2 \left\{1 + \left(\cos\frac{\pi a}{W}\right)^2\right\} + 2\left(\cos\frac{\pi a}{W}\right)^2 - \left(\cos\frac{\pi a}{W}\right)^4\right]$$

$$\overline{Z}_I(z) = \sigma_0 \frac{W}{\pi} \cos\frac{\pi z}{W} \sqrt{\left(\sin\frac{\pi z}{W}\right)^2 - \left(\sin\frac{\pi a}{W}\right)^2} \left[\left(\cos\frac{\pi a}{W}\right)^2 - 2\left(\sin\frac{\pi z}{W}\right)^2\right]$$

$$K_I = \sigma_0 \sqrt{\pi a} \sqrt{\frac{W}{\pi a} \tan\frac{\pi a}{W}} \left(\cos\frac{\pi a}{W}\right)^2 \left[3\left(\cos\frac{\pi a}{W}\right)^2 - 2\right]$$

$$\underset{|x| \le a}{\mathrm{Im}} \left\{\overline{Z}_I(x)\right\} = \frac{\sigma_0 W}{\pi} \cos\frac{\pi x}{W} \sqrt{\left(\sin\frac{\pi a}{W}\right)^2 - \left(\sin\frac{\pi x}{W}\right)^2} \left[\left(\cos\frac{\pi a}{W}\right)^2 - 2\left(\sin\frac{\pi x}{W}\right)^2\right]$$

Crack Opening Area:

$$A = \frac{\sigma_0 W^2}{\pi E'} \left(\sin\frac{\pi a}{W}\right)^2 \left[3\left(\cos\frac{\pi a}{W}\right)^2 - 1\right]$$

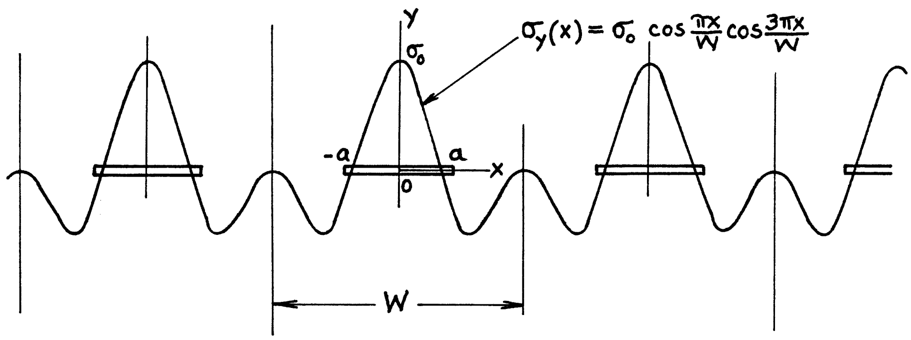
Opening at Center: $\delta = 2v(0, 0)$

$$\delta = \frac{4\sigma_0 W}{\pi E'} \sin\frac{\pi a}{W} \left(\cos\frac{\pi a}{W}\right)^2$$

Relative Vertical Displacement at Infinity due to Crack:

$$\Delta (= \Delta_{crack}) = A/W$$

Method: Westergaard Stress Function (Integration of **page 7.7**)
A, δ, Δ also Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

$$Z_I(z) = \sigma_0 \frac{\sin\frac{\pi z}{W}}{\sqrt{\left(\sin\frac{\pi z}{W}\right)^2 - \left(\sin\frac{\pi a}{W}\right)^2}} \left[4\left(\cos\frac{\pi z}{W}\right)^4 - \left(\cos\frac{\pi z}{W}\right)^2 \left\{1 + 2\left(\cos\frac{\pi a}{W}\right)^2\right\} + \frac{1}{2}\left(\cos\frac{\pi a}{W}\right)^2 \left(\sin\frac{\pi a}{W}\right)^2\right]$$

$$\overline{Z}_I(z) = \sigma_0 \frac{W}{\pi} \cos\frac{\pi z}{W} \sqrt{\left(\sin\frac{\pi z}{W}\right)^2 - \left(\sin\frac{\pi a}{W}\right)^2} \left\{\left(\cos\frac{\pi z}{W}\right)^2 - \frac{1}{2}\left(\sin\frac{\pi a}{W}\right)^2\right\}$$

$$K_I = \sigma_0\sqrt{\pi a}\sqrt{\frac{W}{\pi a}\tan\frac{\pi a}{W}} \cdot \frac{1}{2}\left(\cos\frac{\pi a}{W}\right)^2 \left\{3\left(\cos\frac{\pi a}{W}\right)^2 - 1\right\}$$

$$\operatorname*{Im}_{|x|\le a}\left\{\overline{Z}_I(x)\right\} = \sigma_0 \frac{W}{\pi} \cos\frac{\pi x}{W} \sqrt{\left(\sin\frac{\pi a}{W}\right)^2 - \left(\sin\frac{\pi x}{W}\right)^2} \left\{\left(\cos\frac{\pi x}{W}\right)^2 - \frac{1}{2}\left(\sin\frac{\pi a}{W}\right)^2\right\}$$

Crack Opening Area:

$$A = \frac{\sigma_0 W^2}{\pi E'} \cdot \frac{1}{2}\left(\sin\frac{\pi a}{W}\right)^2 \left[3\left(\cos\frac{\pi a}{W}\right)^2 + 1\right]$$

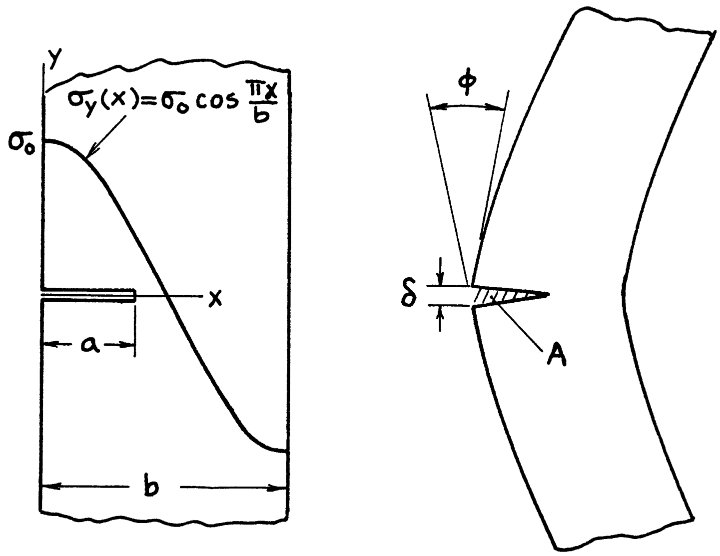
Opening at Center: = $2v(0,0)$

$$\delta = \frac{2\sigma_0 W}{\pi E'} \cdot \sin\frac{\pi a}{W} \left[\left(\cos\frac{\pi a}{W}\right)^2 + 1\right]$$

Relative Vertical Displacement at Infinity due to Crack:

$$\Delta (= \Delta_{crack}) = {}^{A}\!/_{W}$$

Method: Superposition of **pages F.6 and F.7**: $\frac{1}{2}$ **(F.6 + F.7)** (or Integration of **page 7.7**)
A, δ, Δ also Paris' Equation (see **Appendix B**)
Accuracy: Exact
Reference: **Tada 2000**



$$K_I = \sigma_0\sqrt{\pi a}\cdot\frac{1.122 - 1.129\frac{a}{b} + .462\left(\frac{a}{b}\right)^2}{\left(1-\frac{a}{b}\right)^{3/2}}$$

Crack Opening Area:

$$A = \frac{\sigma_0\,\pi a^2}{E'}\cdot\frac{1.258 - 1.62\frac{a}{b} + 1.49\left(\frac{a}{b}\right)^2 - .62\left(\frac{a}{b}\right)^3}{\left(1-\frac{a}{b}\right)^2}$$

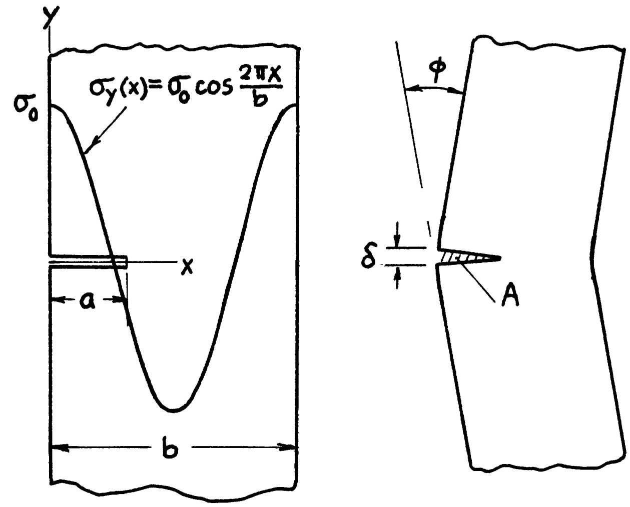
Opening at Edge:

$$\delta = \frac{4\sigma_0\,a}{E'}\cdot\frac{1.164 - .36\frac{a}{b} + .29\left(1-\frac{a}{b}\right)^6}{\left(1-\frac{a}{b}\right)^2}$$

Rotation (Kink) at Cracked Section due to Crack:

$$\phi = \frac{\sigma_0}{E'}\cdot\left(\frac{\frac{a}{b}}{1-\frac{a}{b}}\right)^2\left[23.8 - 40.15\frac{a}{b} + 19.55\left(\frac{a}{b}\right)^2 - \left(15.73 - 11.24\frac{a}{b}\right)\frac{a}{b}\left(1-2\frac{a}{b}\right)\left(1-\frac{a}{b}\right)\right]$$

Methods: K_I Integration of **page 2.27**
A, δ, ϕ Paris' Equation (see **Appendix B**)
Accuracy: K_I 1%; A, δ, ϕ 2%
Reference: **Tada 1985**



$$K_I = \sigma_0\sqrt{\pi a}\left[\frac{1.122 - 4\left(\frac{a}{b}\right)^3}{\left\{1+2\left(\frac{a}{b}\right)^2\right\}^2} + 1.5\frac{a}{b}\left(.375 - \frac{a}{b}\right)\left(1-\frac{a}{b}\right)^3\right]$$

Crack Opening Area:

$$A = \frac{\sigma_0\,\pi a^2}{E'}\cdot\frac{1.258 - .057\frac{a}{b} - 1.76\left(\frac{a}{b}\right)^2}{\sqrt{1-\frac{a}{b}}}$$

Opening at Edge:

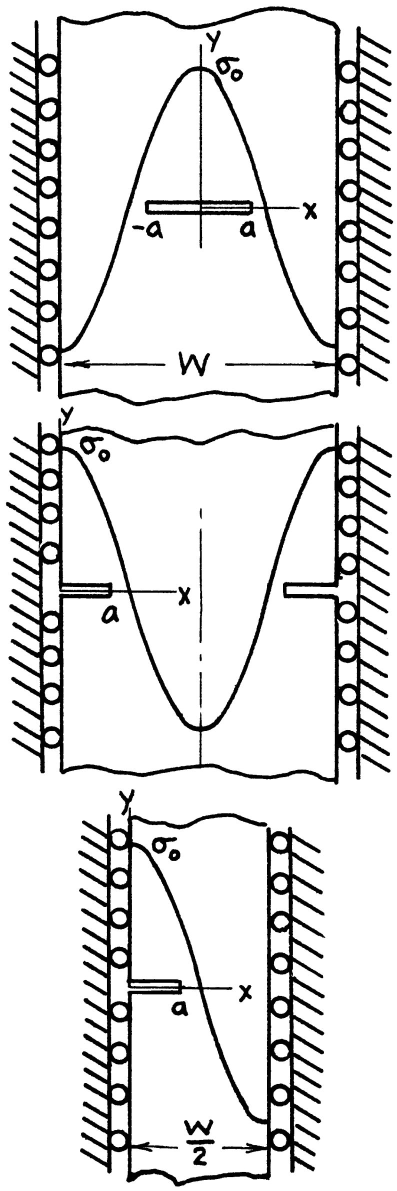
$$\delta = \frac{4\sigma_0\,a}{E'}\cdot\frac{1.454 + .13\frac{a}{b} - 1.72\left(\frac{a}{b}\right)^3 - .76\left(\frac{a}{b}\right)^4}{\sqrt{1-\frac{a}{b}}}$$

Angle of Rotation (Kink at Cracked Section) due to Crack:

$$\phi = \frac{\sigma_0}{E'}\cdot\frac{\left(\frac{a}{b}\right)^2\left(23.8 - 28\frac{a}{b}\right)}{\sqrt{1-\frac{a}{b}}}$$

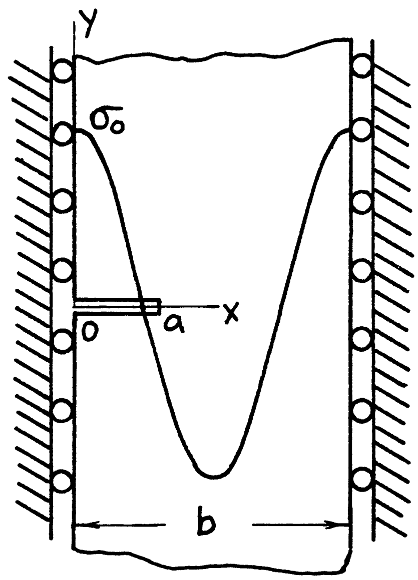
Methods: K_I Integration of **page 2.27**
A, δ, ϕ Paris' Equation (see **Appendix B**)
Accuracy: K_I 1 %; A, δ 2% for $a/b \lesssim 0.65$; ϕ 3% for $a/b \lesssim 0.65$
Reference: **Tada 1985**

NOTE: $K_I < 0$ for $a/b \gtrsim 0.65$



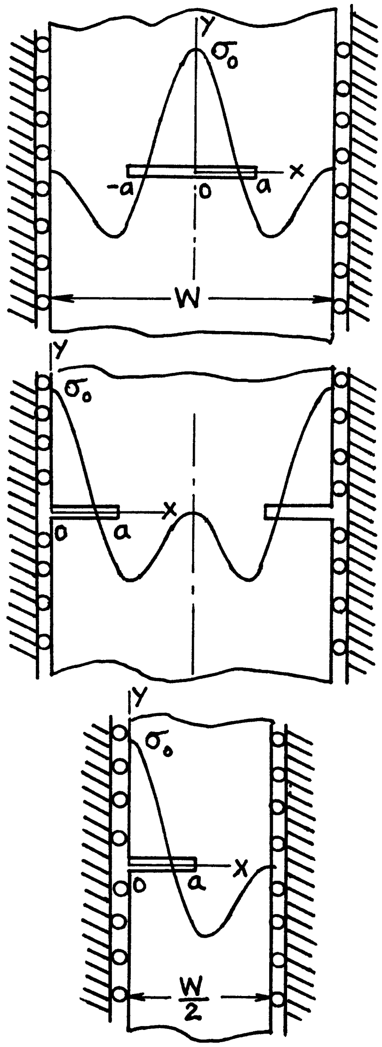
$$\sigma_y(x) = \sigma_0 \cos \frac{2\pi x}{W}$$

See **page F.6**.



$$\sigma_y(x) = \sigma_0 \cos\frac{2\pi x}{b}$$

See **page F.7** with $W = 2b$.

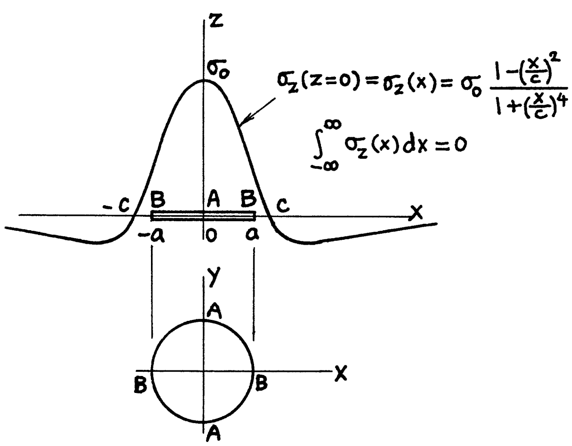


$$\sigma_y(x) = \sigma_0 \cos\frac{\pi x}{W}\cos\frac{3\pi x}{W}$$

See **page F.8**.

$$(K_I)_{\max} = K_{IA} = \frac{2}{\pi}\sigma_0\sqrt{\pi a}\cdot F_A\left(\frac{a}{c}\right)$$

$$(K_I)_{\min} = K_{1B} = \frac{2}{\pi}\sigma_0\sqrt{\pi a}\cdot F_B\left(\frac{a}{c}\right)$$

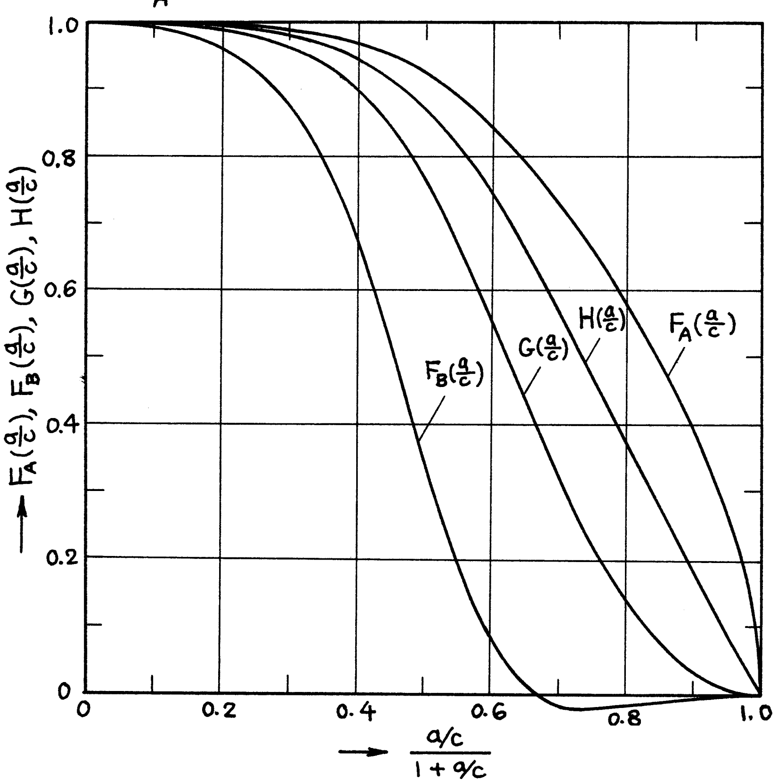


Volume of Crack:

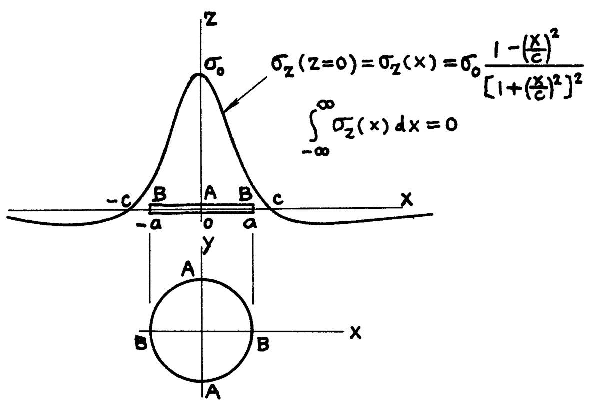
$$V = \frac{16\left(1-\nu^2\right)}{3E}\sigma_0 a^3 \cdot G\left(\frac{a}{c}\right)$$

Opening at Center:

$$\delta = \frac{8\left(1-\nu^2\right)}{\pi E}\sigma_0 a \cdot H\left(\frac{a}{c}\right)$$



Method: Integration of **page 24.11**
Accuracy: Curves are based on numerical values with 0.1% accuracy.
Reference: **Tada 1985**



$$(K_I)_{\max} = K_{IA} = \frac{2}{\pi}\sigma_0\sqrt{\pi a}\cdot F_A\left(\frac{a}{c}\right)$$

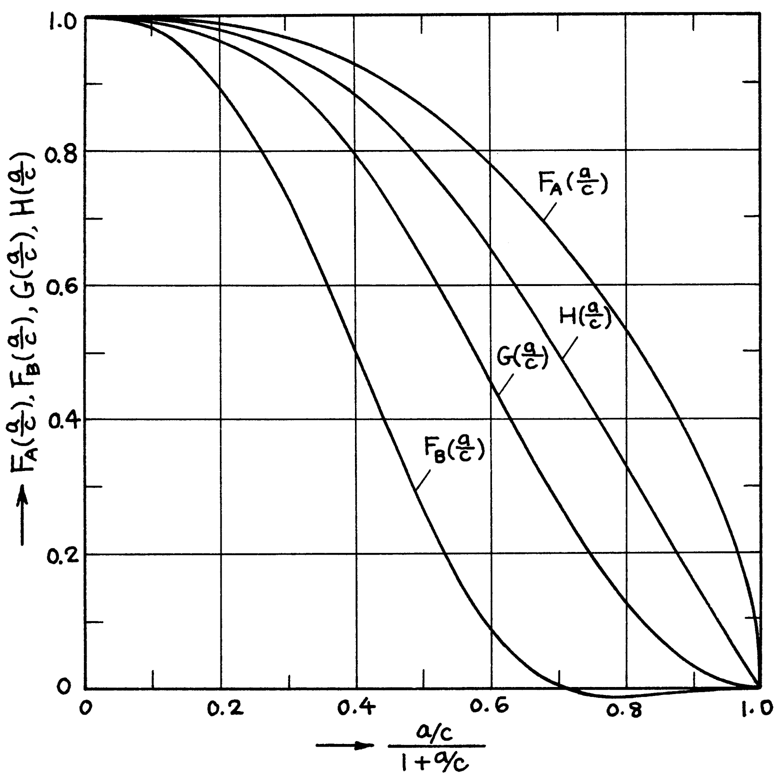
$$(K_I)_{\min} = K_{IB} = \frac{2}{\pi}\sigma_0\sqrt{\pi a}\cdot F_B\left(\frac{a}{c}\right)$$

Volume of Crack:

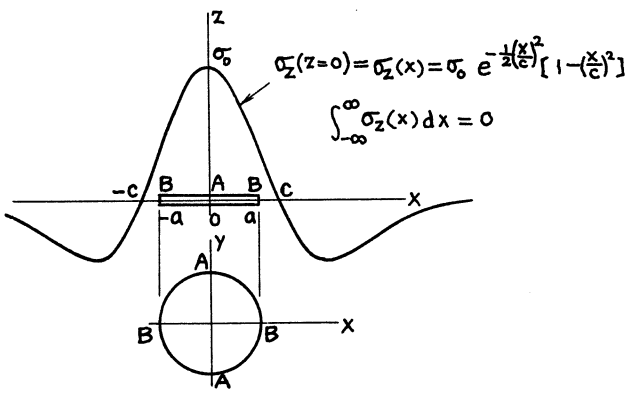
$$V = \frac{16\left(1-\nu^2\right)}{3E}\sigma_0 a^3\cdot G\left(\frac{a}{c}\right)$$

Opening at Center:

$$\delta = \frac{8\left(1-\nu^2\right)}{\pi E}\sigma_0 a\cdot H\left(\frac{a}{c}\right)$$



Method: Integration of **page 24.11**
Accuracy: Curves are based on numerical values with 0.1% accuracy.
Reference: **Tada 1985**



$$(K_I)_{\max} = K_{IA} = \frac{2}{\pi}\sigma_0\sqrt{\pi a}\cdot F_A\left(\frac{a}{c}\right)$$

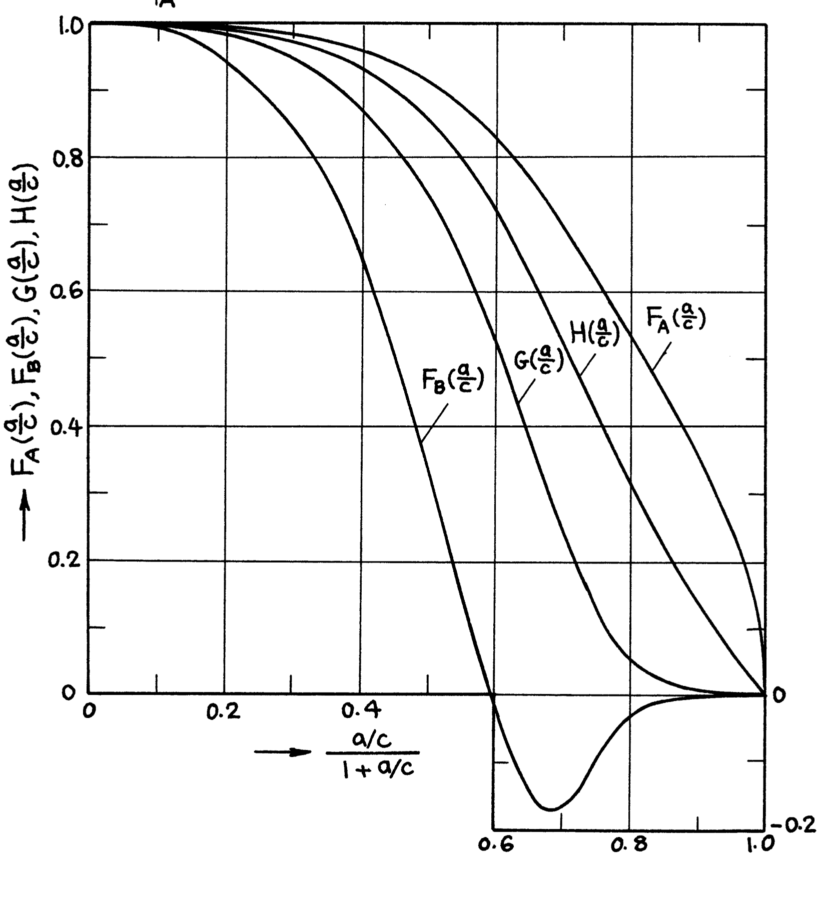
$$(K_I)_{\min} = K_{IB} = \frac{2}{\pi}\sigma_0\sqrt{\pi a}\cdot F_B\left(\frac{a}{c}\right)$$

Volume of Crack:

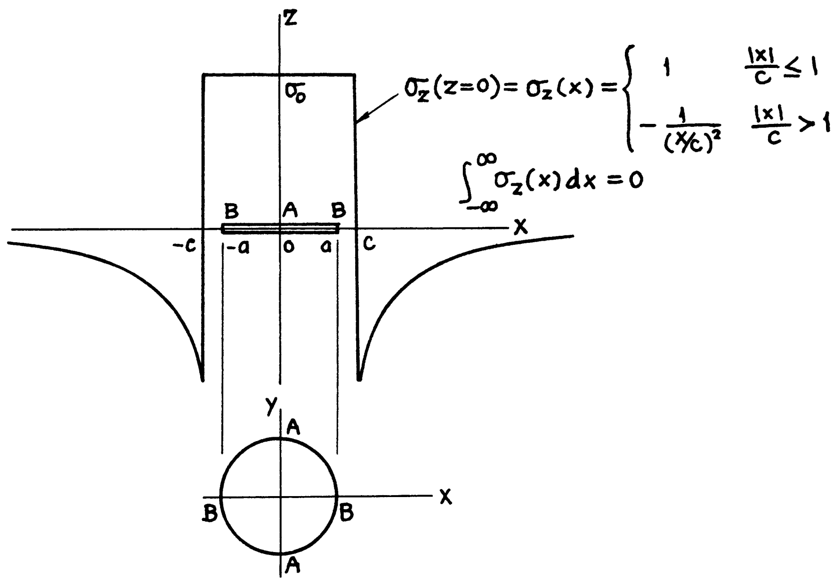
$$V = \frac{16\left(1-\nu^2\right)}{3E}\sigma_0 a^3\cdot G\left(\frac{a}{c}\right)$$

Opening at Center:

$$\delta = \frac{8\left(1-\nu^2\right)}{\pi E}\sigma_0 a\cdot H\left(\frac{a}{c}\right)$$



Method: Integration of **page 24.11**
Accuracy: Curves are based on numerical values with 0.1% accuracy.
Reference: **Tada 1985**



$$(K_I)_{\max} = K_{IA} = \frac{2}{\pi}\sigma_0\sqrt{\pi a}\cdot\begin{cases} 1 & \frac{a}{c} \le 1 \\ \frac{1}{3}\left[4\sqrt{\frac{c}{a}} - \left(\frac{c}{a}\right)^2\right] & \frac{a}{c} > 1 \end{cases}$$

$$(K_I)_{\min} = K_{IB} = \frac{2}{\pi}\sigma_0\sqrt{\pi a}\cdot\begin{cases} 1 & \frac{a}{c} \le 1 \\ F_B\left(\frac{a}{c}\right) & \frac{a}{c} > 1 \end{cases}$$

where

$$F_B\left(\frac{a}{c}\right) = \frac{1}{\sqrt{2}}\left[\left(2-\frac{c}{a}\right)\sqrt{1+\frac{c}{a}} - \left(2+\frac{c}{a}\right)\sqrt{1-\frac{c}{a}} - \left(\frac{c}{a}\right)^2\left(\coth^{-1}\sqrt{2} - \coth^{-1}\sqrt{1+\frac{c}{a}} + \tanh^{-1}\sqrt{1-\frac{c}{a}}\right)\right]$$

Volume of Crack:

$$V = \frac{16\left(1-\nu^2\right)}{3E}\sigma_0 a^3\cdot\begin{cases} 1 & \frac{a}{c} \le 1 \\ \left(\frac{c}{a}\right)^2\left(3 - 2\frac{c}{a}\right) & \frac{a}{c} > 1 \end{cases}$$

Opening at Center:

$$\delta = \frac{8\left(1-\nu^2\right)}{\pi E}\sigma_0 a\cdot\begin{cases} 1 & \frac{a}{c} \le 1 \\ \frac{c}{a}\left(2 - \frac{c}{a}\right) & \frac{a}{c} > 1 \end{cases}$$

Method: Integration of **page 24.11**
Accuracy: Exact
Reference: **Tada 1985**

Appendix G

Westergaard Stress Functions for Dislocations and Cracks[1]

INTRODUCTION

The single-stress function approach of **Westergaard (1939)** is effective for a certain class of crack problems in which cracks occupy collinear line segments (e.g., along the x-axis). Considerable effort has been expended on finding Westergaard stress functions for various load-geometry configurations, and solutions have been obtained for many crack problems (**Westergaard, 1939**). Few Westergaard functions, however, are available for displacement-prescribed problems because of the presumably inherent mathematical difficulties. As noted in a few examples that were solved by other methods, the so-called "rigid wedge" problems or dislocation problems, where displacements of the crack surfaces are partially or wholly prescribed, seem to entail a great deal of mathematical complication involving, in general, single or multiple integral equations (**Barenblatt, 1962; Sneddon, 1969a; Tweed, 1970**).

In this appendix, for ease of subsequent descriptions, the term "dislocations" is generally used in the following sense to distinguish it from the term "cracks": "dislocations" represent macromechanical "cracks with wholly prescribed surface displacements"; whereas "cracks" retain their ordinary interpretation as "cracks with wholly prescribed surface tractions." The micromechanical analogy of "dislocations" will prove to be physically and analytically evident.

The objective of this appendix is to present a different approach to such dislocation problems and then extend it to combined problems of dislocations and cracks using the Westergaard stress function method. It will be shown that the stress functions for dislocations can be more easily obtained than those for cracks and also that the combined problems of dislocations and cracks are nothing more than the ordinary "crack" problems.

The nature of displacement discontinuity and the resulting internal stress field due to a Volterra dislocation in an elastic medium are well known (**Love, 1944; Cottrell, 1953**). The three fundamental displacement discontinuities of this kind are of the form

$$[u] = \Delta_{II}, \quad [v] = \Delta_I, \quad \text{and } [w] = \Delta_{III} \quad \text{on } x < 0, \; y = 0$$

where u, v, and w are, respectively, displacement components in x, y, and z directions, and Δ_{II}, Δ_I, and Δ_{III} are constants. These two-dimensional displacement fields correspond to the Mode II, Mode I, and Mode III displacements, respectively, in fracture mechanics. An analysis based on the theory of dislocation in crystals using distributed screw (Mode III) or edge (Modes I and II) dislocations has also received previous attention (**Cottrell, 1953; Bilby, 1968**). In this appendix, our attention is focused on Mode I problems.

[1] **Published in two parts with some modifications: Tada 1993, and 1994.**

A crack that is filled with a wedge of uniform thickness, the classical dislocation problem, leads to a Westergaard stress function having a $1/z$ singularity at the crack tip or at the end of the wedge. This observation readily leads to the Green's function (i.e., the solution for the "concentrated displacement" problem) for dislocation problems. It is then only necessary to integrate the Green's function to obtain Westergaard stress functions for dislocations with arbitrarily prescribed displacements. Once this method is established, the combined problems of dislocations and cracks are, in effect, reduced to the ordinary crack problems. The present method thus requires no more than integrations, even for the combined problems, and significantly simplifies the analysis.

An analysis of dislocations is presented in the first half of this appendix, which is followed by an analysis of combined problems in the second half. Many examples follow both analyses, and some of them are discussed in detail.

STRESS FUNCTIONS FOR DISLOCATIONS WITH UNIFORM THICKNESS

We define the displacement among the x-axis by

$$D(x) = 2v(x, 0) = v^{+}(x,\ 0) - v^{-}(x,\ 0) \tag{G1}$$

where the $\pm$ signs indicate the upper and lower surfaces along $y = 0$.

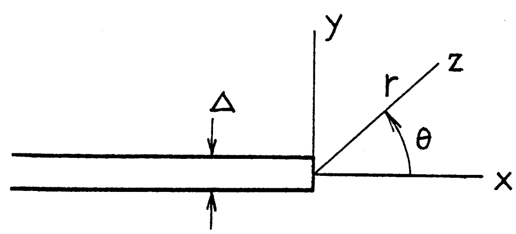


Fig. G1

First we consider the Volterra dislocation with discontinuity along the negative x-axis, as shown in **Fig. G1**. The discontinuous displacement is given by

$$\begin{aligned} D_0(x) &= \Delta\,[1 - H(x)] \\ \text{or}\quad &= \Delta\, H(-x) \end{aligned} \tag{G2}$$

where $H(x)$ denotes the unit step function.

For Westergaard stress functions, $Z(z)$, $z = x + iy$, the displacement along $y = 0$ is given by **Eq. (39)**:

$$2v(x, 0) = \frac{4}{E'}\ \mathrm{Im}\{\,\overline{Z}(z)\}_{y=0} \tag{G3}$$

where the notation Im{ } and, for subsequent use, Re{ } denote the imaginary and real part of { }, respectively, and $\overline{Z}(z) = \int Z(z)dz$, $E' = E$ for plane stress condition, and $E' = E/(1-\nu^2)$ for plane strain condition ($E' = 4\alpha G$). Noting that for polar coordinates (r,θ)

$$\ell\text{n}\, z = \ell\text{n}\, r + i\theta$$

and that $x > 0,\ y = 0$ and $x < 0,\ y = 0$ give $\theta = 0$ and $\theta = \pm\pi$, respectively, $\overline{Z}(z)$, which is in agreement with the displacement condition of **Eq. (G2)** is given by

$$\overline{Z}_0(z) = \frac{E'\Delta}{4\pi}\ell\text{n}\ z \tag{G4}$$

The corresponding Westergaard stress function, $Z(z)$, is

$$Z_0(z) = \frac{E'\Delta}{4\pi}\frac{1}{z} \tag{G4a}$$

Eqs. (G4) and (G4a) are well-known functions for the Volterra dislocation, or for an edge dislocation with the Burgers vector Δ in the y-direction.

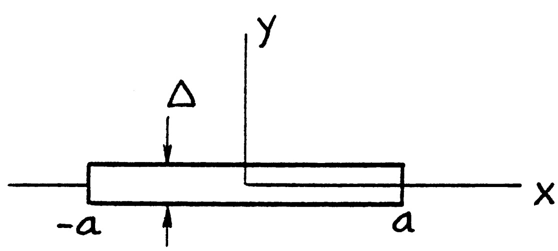


Fig. G2

Next consider a few simple examples that are directly derived from the previous solution. The first example is a uniform dislocation of finite length, as illustrated by **Fig. G2**. Since the displacement along the x-axis is given by

$$\begin{aligned} D(x) &= \Delta\,[H\{-(x-a)\} - H\{-(x+a)\}] \\ \text{or}\quad &= D_0(x-a) - D_0(x+a) \end{aligned} \tag{G5}$$

we can readily write the stress function $Z(z)$ and $\overline{Z}(z)$ by superposition of two dislocations with opposite signs at $(\pm a, 0)$:

$$\begin{aligned} Z(z) &= Z_0(z-a) - Z_0(z+a) \\ &= \frac{E'\Delta}{4\pi}\left(\frac{1}{z-a} - \frac{1}{z+a}\right) \\ \text{or}\quad &= \frac{E'\Delta}{4\pi}\left(\frac{2a}{z^2-a^2}\right) \end{aligned} \tag{G6a}$$

and

$$\overline{Z}(z) = \frac{E'\Delta}{4\pi}\ \ell\text{n}\ \left(\frac{z-a}{z+a}\right) \tag{G6b}$$

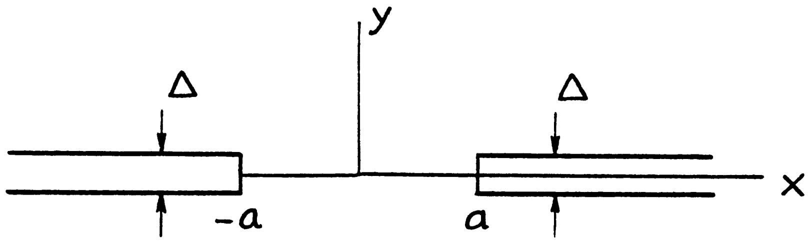


Fig. G3

Similarly, for two semi-infinite dislocations, as shown in Fig. G3,

$$\begin{aligned} D(x) &= \Delta[H\{-(x+a)\} - H(x-a)] \\ \text{or} \quad &= D_0(x+a) - D_0(a-x) \end{aligned} \tag{G7}$$

thus

$$Z(z) = \frac{E'\Delta}{4\pi}\left(\frac{2a}{a^2 - z^2}\right) \tag{G8a}$$

and

$$\overline{Z}(z) = \frac{E'\Delta}{4\pi}\,\ell n\left(\frac{a-z}{a+z}\right) \tag{G8b}$$

For special dislocation problems under consideration, unlike usual stress-prescribed crack problems, there are no interactions (couplings) among stress functions and the stress functions are obtained by direct superposition of individual stress functions for each dislocation that exists independently. Thus for collinearly located dislocations, as shown in **Fig. G4**, since displacement discontinuities are given by

$$D(x) = \sum_{k=1}^{n} \Delta_k [H\{-(x-\beta_k)\} - H\{-(x-\alpha_k)\}] \tag{G9}$$

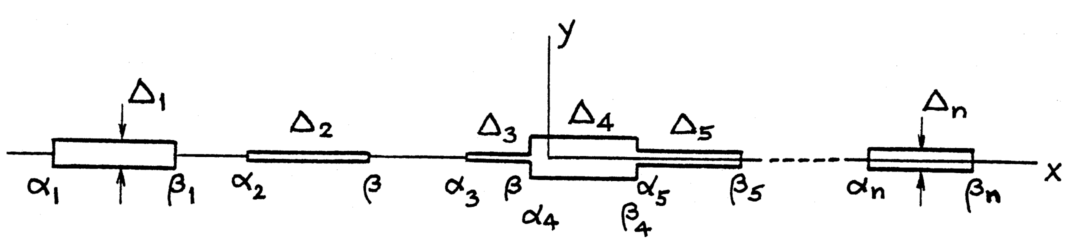


Fig. G4

we can write the stress function $Z(z)$ and $\overline{Z}(z)$ as

$$Z(z) = \frac{E'}{4\pi} \sum_{k=1}^{n} \Delta_k \left[\frac{1}{z-\beta_k} - \frac{1}{z-\alpha_k}\right] \tag{G10a}$$

and

$$\overline{Z}(z) = \frac{E'}{4\pi} \sum_{k=1}^{n} \Delta_k \; \ell\text{n} \left(\frac{z-\beta_k}{z-\alpha_k}\right) \tag{G10b}$$

The situations shown in **Figs. G1, G2, and G3** are special cases of **Fig. G4**. In **Fig. G4**, if we put, for example, $\Delta_1 = \Delta$, $\alpha_1 = -\infty$, $\beta_1 = 0$ with $\Delta_k = 0(k = 2, \cdots, n)$, $\Delta_1 = \Delta$, $\alpha_1 = -a$, $\beta_1 = a$ with $\Delta_k = 0$ $(k = 2, \cdots, n)$, and $\Delta_1 = \Delta_n = \Delta$, $\alpha_1 = -\infty$, $\beta_1 = -a$, $\alpha_n = a$, $\beta_n = +\infty$ with $\Delta_k = 0$ $(k = 2, \cdots, n-1)$, we have the situations of **Figs. G1, G2, and G3**, respectively.

Another example is the periodic array of identical dislocations with discontinuity Δ, length $2a$, and period $2b$, as shown in **Fig. G5**.

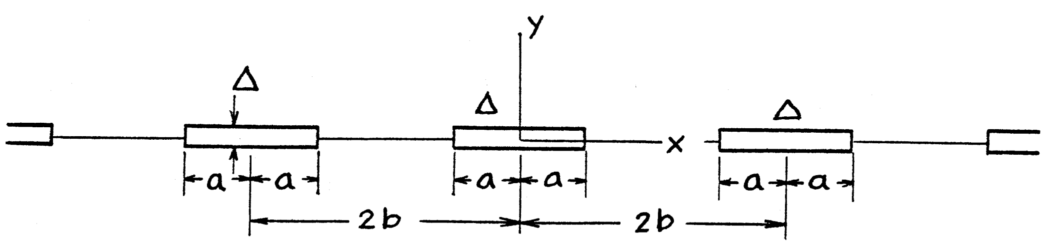


Fig. G5

Substituting $\Delta_k = \Delta$, $\alpha_k = 2kb - a$, $\beta_k = 2kb + a$ into **Eqs. (G10a) and (G10b)**, we can write $Z(z)$ and $\overline{Z}(z)$ as

$$Z(z) = \frac{E'\Delta}{4\pi} \sum_{k=-\infty}^{\infty} \left(\frac{1}{z-a-2kb} - \frac{1}{z+a-2kb}\right) \tag{G11a}$$

$$\overline{Z}(z) = \frac{E'\Delta}{4\pi} \sum_{k=-\infty}^{\infty} \ell\text{n} \left(\frac{z-a-2kb}{z+a-2kb}\right) \tag{G11b}$$

After some manipulations, we apply the identities given by **Eqs. (G12a) and (G12b)** to **Eqs. (G11a) and (G11b)**, respectively.

$$\sum_{k=0}^{\infty} \frac{1}{k^2-\alpha^2} = -\frac{\pi}{2\alpha}\left[\frac{1}{\pi\alpha} + \cot(\pi\alpha)\right] \tag{G12a}$$

$$\prod_{k=0}^{\infty} \left(1 - \frac{\alpha^2}{k^2}\right) = \frac{\sin\ \pi\alpha}{\pi\alpha} \tag{G12b}$$

Then these functions are reduced to the following simple, closed-form functions:

$$Z(z) = \frac{E'\Delta}{4\pi}\frac{\pi}{2b}\left[\cot\frac{\pi(z-a)}{2b} - \cot\frac{\pi(z+a)}{2b}\right] \qquad \textbf{(G13a)}$$

$$\overline{Z}(z) = \frac{E'\Delta}{4\pi}\ \ell\text{n}\ \frac{\sin\frac{\pi(z-a)}{2b}}{\sin\frac{\pi(z+a)}{2b}} \qquad \textbf{(G13b)}$$

Note that either **Eq. (G13a) or (G13b)** can be derived directly from the other by the relationship $\overline{Z}(z) = \int Z(z)dz$ or $Z(z) = d\overline{Z}(z)/dz$.

GREEN'S FUNCTIONS FOR DISLOCATIONS OF ARBITRARY SHAPE

The previous results suggest the possibility of constructing stress functions for dislocations of arbitrary shape. We have considered so far the elastic response to dislocations whose discontinuous displacements are represented by the sum of the unit step functions. In order to obtain elastic solutions for dislocation problems of arbitrary displacement, it is more convenient to consider the elastic response to the dislocation displacement in the form of the unit impulse or the delta function. The delta function, $\delta(x)$, is related to the unit step function, $H(x)$, as follows:

$$\delta(x) = \frac{dH(x)}{dx} = -\frac{dH(-x)}{dx} \qquad \textbf{(G14)}$$

The same relationship is satisfied between the elastic responses to the unit step dislocation, $H(x):\ Z_H(z)$, $\overline{Z}_H(z)$, and those to the unit impulse dislocation, $\delta(x):\ Z_\delta(z),\ \overline{Z}_\delta(z)$.

$$Z_\delta(z) = \frac{dZ_H(z)}{dx} = \frac{dZ_H(z)}{dz} \qquad \textbf{(G15a)}$$

$$\overline{Z}_\delta(z) = \frac{d\overline{Z}_H(z)}{dz} = Z_H(z) \qquad \textbf{(G15b)}$$

Comparison of **Eqs. (G14), (G15a), and (G15b)** with **Eqs. (G2), (G4a), and (G4)** leads to the stress functions for the dislocation with displacement density $d \cdot \delta(x)$. This nucleus of displacement may be construed as a "concentrated displacement". Noting that $Z_H(z) = -\frac{1}{\Delta}Z_0(z)$, these functions are written as follows:

$$Z(z) = -\frac{d}{\Delta}Z_0'(z) = \frac{E'd}{4\pi}\frac{1}{z^2} \qquad \textbf{(G16a)}$$

$$\overline{Z}(z) = -\frac{d}{\Delta}Z_0(z) = -\frac{E'd}{4\pi}\frac{1}{z} \qquad \textbf{(G16b)}$$

These functions serve as the Green's functions for dislocations of arbitrary shape.

The distinct advantage of this method is that the desired displacements can be built up by putting together elementary "concentrated displacements" with no cracks to begin with, thus avoiding the interactions (couplings) among the solutions for individual dislocations. That is, these solutions can be directly superimposed; consequently, Westergaard stress functions are, in general, actually more readily obtained for dislocations than for cracks.

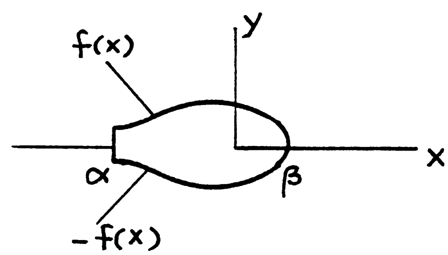


Fig. G6

Let a dislocation displacement on the x-axis, $D(x)$, defined by **Eq. (G1)**, be given by a piecewise continuous function $f(x)$ that is nonnegative[2] in a region $\alpha \le x \le \beta$ and zero in regions $x < \alpha$ and $x > \beta$, as shown in **Fig. G6**. That is,

$$\begin{aligned} D(x) &= 2f(x), \qquad && \alpha \le x \le \beta \\ D(x) &= 0 && x < \alpha,\ x > \beta \end{aligned} \tag{G17}$$

Then, using the Green's functions given by **Eqs. (G16a) and (G16b)**, the Westergaard stress functions $Z(z)$ and $\overline{Z}(z)$ are obtained by simple integrations.

$$Z(z) = \frac{E'}{2\pi} \int_{\alpha}^{\beta} \frac{f(x')}{(z - x')^2} dx' \tag{G18a}$$

$$\overline{Z}(z) = -\frac{E'}{2\pi} \int_{\alpha}^{\beta} \frac{f(x')}{z - x'} dx' \tag{G18b}$$

Again, note that either $Z(z)$ or $\overline{Z}(z)$ of **Eq. (G18)** is also derived directly from the other.

[2] **This requirement is only for physical purposes.**

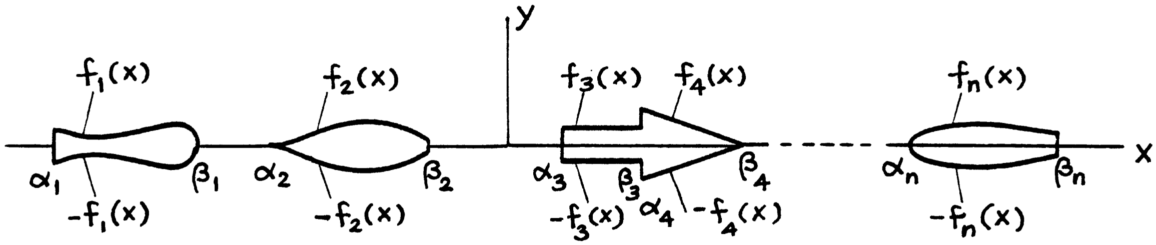


Fig. G7

When the dislocation displacement, $D(x)$, consists of more than one piecewise continuous functions, $2f_i(x)$, which are nonnegative in $\alpha_i \leq x \leq \beta_i$ and zero in $\beta_{i-1} < x < \alpha_i$ and $\beta_i < x < \alpha_{i+1}$, as shown in **Fig. G7**, by simply applying the direct superposition of stress functions, we can write the stress functions as

$$Z(z) = \frac{E'}{2\pi}\sum_{i=1}^{n}\int_{\alpha_i}^{\beta_i}\frac{f_i(x')}{(z-x')^2}dx' \qquad \textbf{(G19a)}$$

$$\overline{Z}(z) = -\frac{E'}{2\pi}\sum_{i=1}^{n}\int_{\alpha_i}^{\beta_i}\frac{f_i(x')}{z-x'}dx' \qquad \textbf{(G19b)}$$

Note that integrating **Eq. (G18a)** by parts we have its alternative form

$$Z(z) = \frac{E'}{2\pi}\left[\frac{f(x')}{z-x'}\bigg|_{\alpha}^{\beta} - \int_{\alpha}^{\beta}\frac{f'(x')}{z-x'}dx'\right] \qquad \textbf{(G18a)}'$$

where $f'(x) = df(x)/dx$. The choice between **Eqs. (G18a) and (G18a)**$'$ can be made on the simplicity of the integration.

WESTERGAARD STRESS FUNCTIONS FOR SEVERAL DISLOCATION PROBLEMS

Some examples are given to illustrate the possibilities for easily obtaining solutions.

1. Periodically located identical dislocations whose displacements densities are given by $d \cdot \delta(x \pm 2kb)$, $k = 0, \ldots, \infty$ (**Fig. G8**).

Applying the relations given by the first halves of **Eqs. (G16a) and (G16b)** to **Eqs. (G13a) and (G13b)**, respectively, we readily have

$$Z(z) = \frac{E'd}{4\pi}\left(\frac{\pi}{2b}\operatorname{cosec}\frac{\pi z}{2b}\right)^2 \qquad \textbf{(G20a)}$$

$$\overline{Z}(z) = \frac{E'd}{4\pi}\left(-\frac{\pi}{2b}\cot\frac{\pi z}{2b}\right) \qquad \textbf{(G20b)}$$

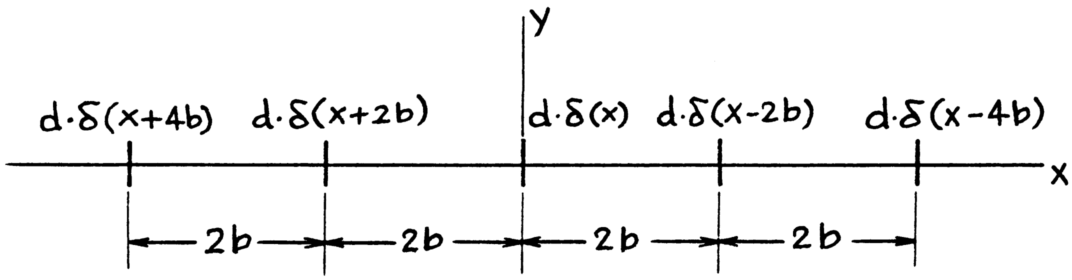


Fig. G8

These functions can also be obtained from the Green's functions, **Eqs. (G16a) and (G16b)**, by the direct superposition

$$Z(z) = \frac{E'd}{4\pi} \sum_{k=-\infty}^{\infty} \frac{1}{(z-2kb)^2}$$

$$\overline{Z}(z) = -\frac{E'd}{4\pi} \sum_{k=-\infty}^{\infty} \frac{1}{z-2kb}$$

with the aid of the following identity [or the relation $\overline{Z}(z) = \int Z(z)dz$] in addition to **Eq. (G12a)**

$$\sum_{k=0}^{\infty} \frac{a^2+k^2}{\left(k^2-a^2\right)^2} = \frac{1}{2}\left[\frac{1}{a^2} + (\pi \operatorname{cosec} \pi a)^2\right] \qquad \text{(G12c)}$$

Eqs. (G20a) and (G20b) can serve as the Green's functions for periodically repeated dislocations with arbitrary shape.

2a. Diamond-shaped dislocation (**Fig. G9**).
$f(x)$ is given by

$$f(x) = \frac{\Delta_0}{2}\left(1 - \frac{|x|}{2}\right)$$

or

$$= \alpha(a - |x|) \qquad |x| \le a$$

$$f(x) = 0 \qquad |x| > a$$

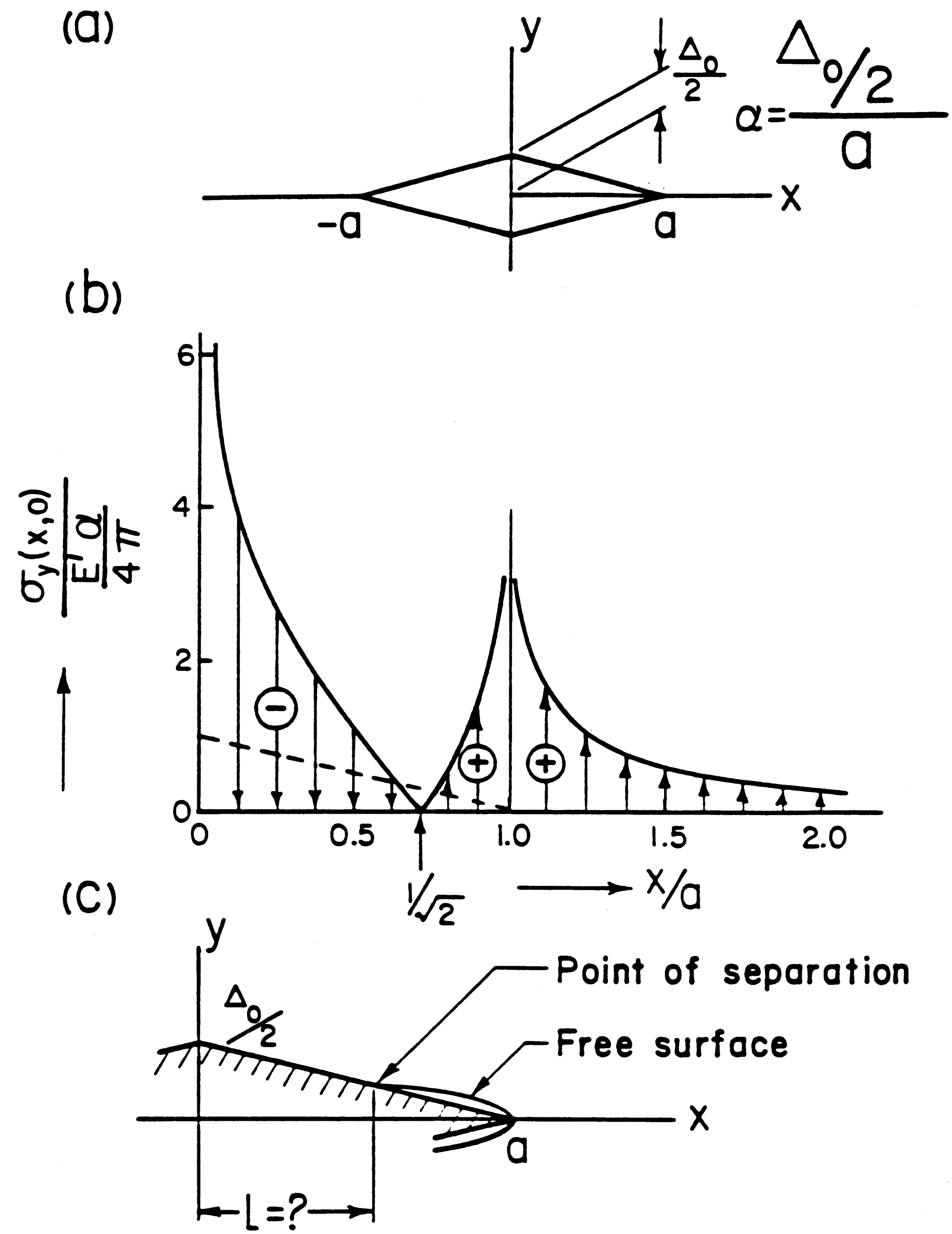

(a)
y
$\frac{\Delta_0}{2}$
$\alpha = \frac{\Delta_0/2}{a}$
x
-a
a
(b)
$\frac{\sigma_y(x,0)}{\frac{E'\alpha}{4\pi}}$
6
4
2
0
0
0.5
1.0
1.5
2.0
$1/\sqrt{2}$
x/a
(c)
y
$\Delta_0/2$
Point of separation
Free surface
x
a
L=?

Fig. G9

Using **Eq. (G18a)**, we have the following Westergaard stress function:

$$Z(z) = \frac{E'\alpha}{4\pi}\ \ell\text{n}\ \frac{z^2}{(z+a)(z-a)} \tag{G21a}$$

and

$$\begin{aligned}\overline{Z}(z) &= \int Z(z)dz \\ &= \frac{E'\alpha}{4\pi}\ \ell\text{n}\ \frac{\left(z^2\right)^z}{(z+a)^{z+a}(z-a)^{z-a}}\end{aligned} \tag{G21b}$$

The following observation supports visualization of our intuition that we cannot always fill a crack with a rigid wedge of arbitrary shape without welding the surfaces together.

The normal stress distribution, $\sigma_y(x, 0)$, is calculated from **Eq. (G21a)** using **Eq. (38)**

$$\begin{aligned}\sigma_y(x,0) &= \text{Re}\{Z(z)\}_{y=0} \\ &= \frac{E'\alpha}{4\pi}\ \ell\text{n}\ \frac{x^2}{(x+a)(x-a)},\quad |x|> a \\ &= \frac{E'\alpha}{4\pi}\ \ell\text{n}\ \frac{x^2}{(a+x)(a-x)},\quad |x|< a\end{aligned} \tag{G21a$'$}$$

The stress distribution given by **Eq. (G21a)$'$** is plotted in **Fig. G9b**. As readily seen, to maintain the diamond shape we have to apply the tensile surface stress over the region $a/\sqrt{2} < x < a$. This implies that when a diamond-shaped wedge is inserted into a crack with equal length, the contact between the surfaces is lost and free surfaces are formed near the tips of the wedge (see **Fig. G9c**). The location of the point of separation is, however, not known beforehand. The method for determining the position of this point is discussed subsequently.

2b. Dislocation shown in **Fig. G10**.

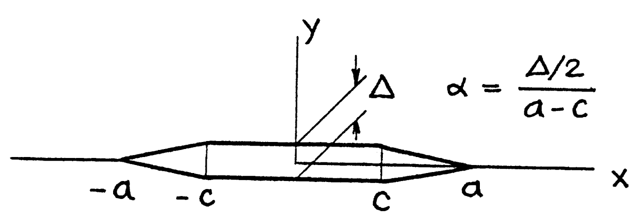


Fig. G10

$$f(x) = \Delta/2 = \alpha(a-c) \qquad |x| \le c$$

$$f(x) = \alpha(a-|x|) \qquad c \le |x| \le a$$

$$f(x) = 0 \qquad |x| > a$$

By direct superpositions of **Eqs. (G21a) and (G21b)**, we immediately have the following solutions:

$$Z(z) = \frac{E'\alpha}{4\pi}\ \ell\text{n}\ \frac{(z+c)(z-c)}{(z+a)(z-a)} \tag{G22a}$$

$$\overline{Z}(z) = \frac{E'\alpha}{4\pi}\ \ell\text{n}\ \frac{(z+c)^{z+c}(z-c)^{z-c}}{(z+a)^{z+a}(z-a)^{z-a}} \tag{G22b}$$

3a. Periodically located diamond-shaped dislocations (**Fig. G11**).

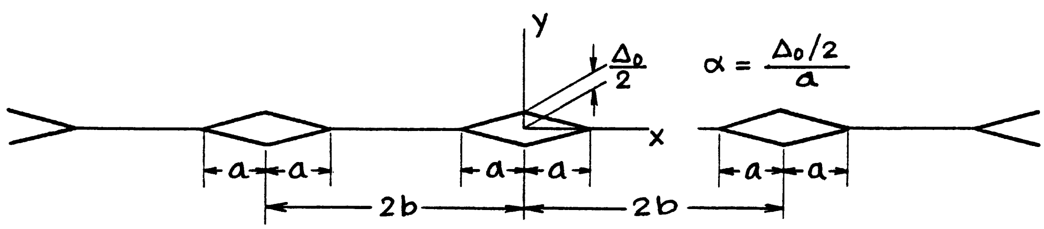


Fig. G11

By superposition of **Eq. (G21a)**, $Z(z)$ is given by

$$Z(z) = \frac{E'\alpha}{4\pi} \sum_{k=-\infty}^{\infty} F(z-2kb)$$

where

$$F(z) = \ \ell\text{n}\ \frac{z^2}{(z+a)(z-a)}$$

Using the identity given by **Eq. (G12b)**, we have $Z(z)$ in the following simple closed form:

$$Z(z) = \frac{E'\alpha}{4\pi}\ \ell\text{n}\ \frac{\left(\sin\frac{\pi z}{2b}\right)^2}{\sin\frac{\pi(z+a)}{2b}\sin\frac{\pi(z-a)}{2b}} \tag{G23}$$

3b. Periodic dislocations shown in **Fig. G12**.

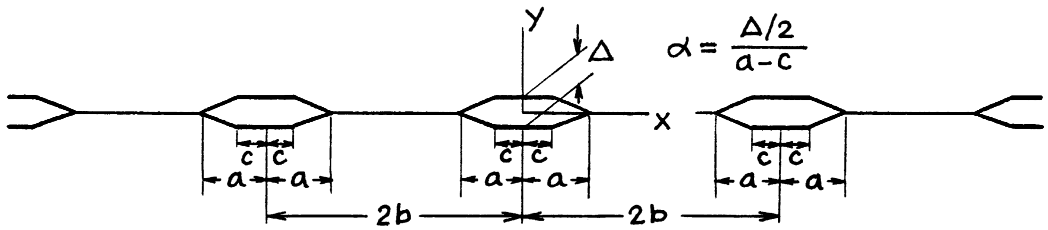


Fig. G12

Directly from **Eq. (G23)**, we have

$$Z(z) = \frac{E'\alpha}{4\pi}\,\ell\text{n}\,\frac{\sin\frac{\pi(z+c)}{2b}\,\sin\frac{\pi(z-c)}{2b}}{\sin\frac{\pi(z+a)}{2b}\,\sin\frac{\pi(z-a)}{2b}} \tag{G24}$$

or

$$= \frac{E'\alpha}{4\pi}\,\ell\text{n}\,\frac{\left(\sin\frac{\pi z}{2b}\right)^2 - \left(\sin\frac{\pi c}{2b}\right)^2}{\left(\sin\frac{\pi z}{2b}\right)^2 - \left(\sin\frac{\pi a}{2b}\right)^2}$$

4. Elliptical-shaped dislocation (**Fig. G13**).

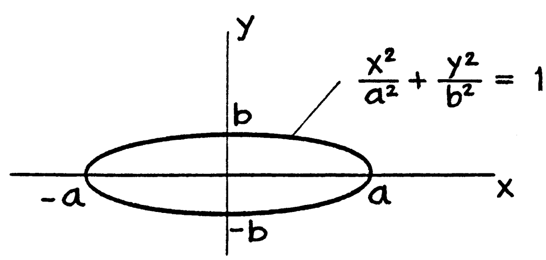


Fig. G13

$f(x)$ for this case is given by

$$f(x) = b\sqrt{1 - \left(\frac{x}{a}\right)^2} \qquad |x| \le a$$

$$f(x) = 0 \qquad |x| > a$$

We can immediately write the stress functions $Z(z)$ and $\overline{Z}(z)$ for this problem by comparison with the well-known solution of the Griffith crack under uniform internal pressure as follows [see **Eq. (71)**].

$$Z(z) = \frac{E'b}{2a}\left[\frac{1}{\sqrt{1-(a/z)^2}} - 1\right] \qquad \text{(G25a)}$$

$$\overline{Z}(z) = \frac{E'b}{2a}\left[\sqrt{z^2 - a^2} - z\right] \qquad \text{(G25b)}$$

In fact, **Eqs. (G25a) and (G25b)** are obtained by the direct application of **Eqs. (G18a) and (G18b)**, respectively.

The solution shows that, when an internal crack in an infinite plate is filled with a smooth rigid wedge with elliptical shape, the traction between the surfaces becomes uniform compression. If uniform remote tensile stress, $E'b/2a$, is applied, these surfaces become traction free.

5. Parabolic-shaped dislocation (**Fig. G14**).

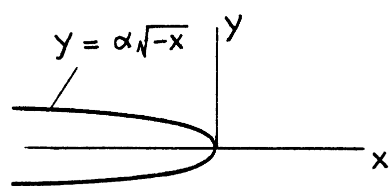


Fig. G14

$f(x)$ for this example is

$$f(x) = \alpha\sqrt{-x} \qquad x \le 0$$

$$f(x) = 0 \qquad x > 0$$

Again we can immediately expect the functions $Z(z)$ and $\overline{Z}(z)$ to be identical to those for a semi-infinite crack with stress-free surfaces along the negative x-axis (see **page 3.1**). In fact, by **Eq. (G18a)**,

$$\begin{aligned} Z(z) &= \frac{E'\alpha}{2\pi}\int_{-\infty}^{0} \frac{\sqrt{-x'}}{(z-x')^2}\,dx' = \frac{E'\alpha}{2\pi}\int_{0}^{\infty} \frac{\sqrt{\xi}}{(z+\xi)^2}\,d\xi \\ &= \frac{E'\alpha}{4}\frac{1}{\sqrt{z}} \end{aligned} \qquad \text{(G26a)}$$

and

$$\overline{Z}(z) = \int Z(z)\,dz = \frac{E'\alpha}{2}\sqrt{z} \qquad \text{(G26b)}$$

Eq. (G26) gives the elastic field near the tip of a crack with free surfaces on $x < 0,\ y = 0$, where the remote loads cause the stress intensity factor $K = \sqrt{\pi}E'\alpha/(2\sqrt{2})$, which is readily obtained by comparing (G26a) with $Z(z) = K/\sqrt{2\pi z}$ on **page 3.1**. The crack-tip field for the preceding example (4), since $K = (E'b/2a)\sqrt{\pi a}$, corresponds to the case where $\alpha = \sqrt{2}\,b/\sqrt{a}$.

6. A Dislocation whose shape is given by **Eq. (G27)** (**Fig. G15**).

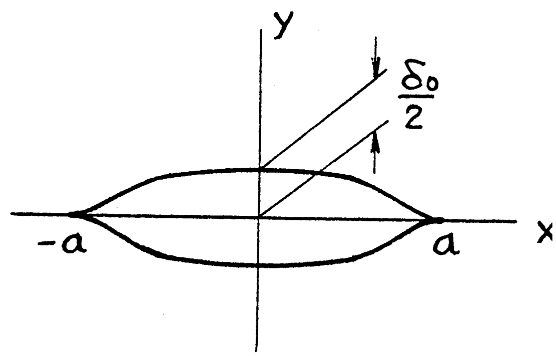


Fig. G15

$$
\begin{aligned}
f(x) &= \frac{\delta_0}{2}\left\{1 - \left(\frac{x}{a}\right)^2\right\}^{\alpha}, \qquad \alpha \geq 0 \qquad |x| \leq a \\
f(x) &= 0 \qquad\qquad\qquad\qquad\qquad\qquad |x| > a
\end{aligned}
\tag{G27}
$$

By **Eq. (G18a)**, $Z(z)$ can be obtained from the following integral:

$$
Z(z) = \frac{E'\delta_0}{4\pi}\int_{-a}^{a}\frac{\left\{1 - \left(\frac{x'}{a}\right)^2\right\}^{\alpha}}{(z - x')^2}dx'
$$

After some manipulations, we arrive at

$$
Z(z) = \frac{E'\delta_0}{2\sqrt{\pi a}}\frac{\Gamma(\alpha + 1)}{\Gamma\left(\alpha + \frac{1}{2}\right)}{}_2F_1\left(1,\ \frac{1}{2} - \alpha;\ \frac{1}{2};\ \left(\frac{z}{a}\right)^2\right)
\tag{G28}
$$

where ${}_2F_1(\alpha,\ \beta;\ \gamma;\ z)$ is the usual notation for the hypergeometric series.

Note that special cases $\alpha = 0$ and $\alpha = \frac{1}{2}$ correspond to the cases of **Fig. G2** and **Fig. G13**, respectively.

Sneddon (1969a) solved this problem as an example of his general solution which was derived by the method of Fourier transforms and dual integral equations. His interpretation of the problem is to find the surface stress distribution that is necessary to maintain the Griffith crack in the prescribed shape. The general solution was given in the form of two relatively simple integrations. The present method requires only one simple integral and seems to provide a simpler and more direct approach to the problem.

The stress distribution, $\sigma_y(x, 0)$, calculated from these stress functions on the dislocations (i.e., the displacement-prescribed segments) is interpreted as the surface stress necessary to maintain the crack surfaces in the prescribed shape. Therefore, when this stress is compressive, that is, $\sigma_y(x, 0) < 0$, over the entire portion of the crack, the problem is equivalent to the insertion of a smooth rigid wedge with prescribed shape into the crack. Otherwise, the stress distribution is regarded as the traction between the surfaces that would result if the surfaces of the wedge and crack are welded together. When the surfaces are not welded, free surfaces form over certain portions of the crack, as discussed in the example (4) above (**Fig. G9**). The positions of the points of separation of the surfaces are determined by the method discussed in the following sections.

COMBINED PROBLEMS OF DISLOCATIONS AND CRACKS

When dislocations (or cracks with fully prescribed displacements) and cracks in the ordinary sense (cracks with prescribed surface stress distributions) are present collinearly on the x-axis, the Westergaard stress functions can be obtained relatively easily. The method is simply successive applications of the preceding method of dislocation analysis and the usual method of stress analysis of cracks. The procedure of the analysis is as follows.

First, obtain the stress function for the dislocation only, $Z_D(z)$, assuming that cracks are not present, by use of the method described in the preceding sections. The normal stress distribution on the x-axis due to the dislocation, $\sigma_{yD}(x, 0)$, is calculated by **Eq. (38)**

$$\sigma_{yD}(x,0) = \mathrm{Re}\{Z_D(z)\}_{y=0} \qquad \textbf{(G29)}$$

This is the "crack-absent" stress distribution to be removed along the cracks. Thus the problem is now reduced to the ordinary crack problem where the crack surfaces are assumed to be stress free. Green's functions for the Westergaard stress function, $Z_G(z, x')$, are available in this handbook for various crack geometries. The stress function for the crack, $Z_C(z)$, under the stress given by **Eq. (G29)**, is calculated by the integral

$$Z_C(z) = \int_C \sigma_{yD}(x', 0) Z_G(z, x') dx' \qquad \textbf{(G30)}$$

where the integral $\int_C (\ldots) dx'$ is a definite integral over the entire portion of the crack. Since the stress function, $Z_C(z)$, generates no vertical displacement, $v(x, 0)$, outside the crack, the prescribed shape of dislocation is maintained. Thus the combined stress function for the dislocation and the crack is obtained by the superposition of $Z_D(z)$ and $Z_C(z)$, that is,

$$Z(z) = Z_D(z) + Z_C(z) \qquad \textbf{(G31)}$$

As discussed earlier, the normal stress, $\sigma_y(x,\ 0)$, on the dislocations, calculated from the total stress function $Z(z)$, **Eq. (G31)**, is the stress distribution necessary to maintain the dislocations in the prescribed shapes. When this stress is compressive over the entire segments of dislocations, the problem is equivalent to plugging the thin rigid wedges with the same prescribed shapes into the corresponding cracks. When the stress is tensile over any portion of the segments, the equivalency is retained only if surfaces of the cracks and the wedges are welded. Otherwise, the contact between the surfaces will be lost and free surfaces will form. Then, the portions of the free surface should be treated as the usual stress-prescribed cracks, that is, $Z_C(z)$ should incorporate the analysis of these portions while $Z_D(z)$ remains unchanged. The points of separation of the surfaces, however, are generally not known beforehand. The method for determining the positions of the separation points is also discussed subsequently.

In the subsequent discussion, the method is applied to several very simple examples to illustrate the possibilities for obtaining the stress function for more complicated problems without great mathematical difficulties.

A CRACK PARTIALLY FILLED WITH A RIGID WEDGE

As a simple illustrative problem of this type, consider a semi-infinite, thin rigid wedge with uniform thickness, Δ, inserted into a semi-infinite crack leaving free surfaces of length $2a$ ahead of the wedge, as shown in **Fig. G16**. Note that **Fig. G16** is a limiting case of **Fig. G23**. For convenience, we take the origin of the coordinates at the center of the stress-free portion.

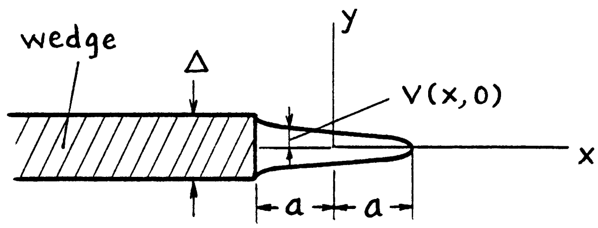


Fig. G16

When there is no free surface ahead of the wedge (or when the crack is filled with the wedge), the stress function is given by **Eq. (G4a)**. That is, the stress function, $Z_D(z)$, in this case is

$$Z_D(z) = \frac{E'\Delta}{4\pi}\frac{1}{z+a} \tag{G32}$$

The corresponding crack-absent stress distribution, $\sigma_{yD}(x,\,0)$, is calculated from **Eq. (G29)**

$$\sigma_{yD}(x,\,0) = \mathrm{Re}\{Z_D(z)\}_{y=0} = \frac{E'\Delta}{4\pi}\frac{1}{x+a} \tag{G33}$$

The Green's function, $Z_G(z,x')$, for a crack in an infinite plane is well known as (see **page 5.10**)

$$Z_G(z,x') = \frac{1}{\pi}\frac{\sqrt{a^2-x'^2}}{(z-x')\sqrt{z^2-a^2}} \tag{G34}$$

The additional stress function due to crack, $Z_C(z)$, is calculated by performing the integration of **Eq. (G30)**

$$\begin{aligned} Z_C(z) &= \int_{-a}^{a} \sigma_{yD}(x',0)Z_G(z,x')dx' \\ &= \frac{E'\Delta}{4\pi^2}\frac{1}{\sqrt{z^2-a^2}}\int_{-a}^{a}\frac{1}{z-x'}\sqrt{\frac{a-x'}{a+x'}}dx' \end{aligned}$$

The final expression of $Z_C(z)$ has the following simple form.

$$Z_C(z) = \frac{E'\Delta}{4\pi}\left\{\frac{1}{\sqrt{z^2-a^2}} - \frac{1}{z+a}\right\} \tag{G35}$$

The total stress function of the problem is then, by **Eq. (G31)**, written as

$$Z(z) = Z_D(z) + Z_C(z) = \frac{E'\Delta}{4\pi}\frac{1}{\sqrt{z^2 - a^2}} \tag{G36a}$$

and

$$\overline{Z}(z) = \int Z(z)dz = \frac{E'\Delta}{4\pi}\cosh^{-1}\frac{z}{a} \tag{G36b}$$

These functions are among the few Westergaard functions known for displacement-prescribed crack problems.

The stress intensity factor, $K_{\pm a}$, at the ends of the free surface, $x = \pm a$, is directly obtained from **Eq. (G36a)** as

$$\begin{aligned} K_{+a} &= \lim_{x\to a+0}\left\{\sqrt{2\pi(x-a)}\ \sigma_y(x,0)\right\} \\ &= \lim_{x\to a+0}\left[\sqrt{2\pi(x-a)}\ \ \mathrm{Re}\{Z(z)\}_{y=0}\right] \\ K_{+a} &= \frac{E'\Delta}{4\sqrt{\pi a}} \end{aligned} \tag{G37a}$$

Similarly,

$$K_{-a} = -\frac{E'\Delta}{4\sqrt{\pi a}} \tag{G37b}$$

The profile of the free surface (see **Fig. G16**), $v(x,0)$; $|x| \le a$, is calculated from **Eq. (G36b)** by the formula given by **Eq. (G3)**.

$$\begin{aligned} \underset{|x|\le a}{v(x,0)} &= \frac{2}{E'}\ \ \mathrm{Im}\{\overline{Z}(z)\}_{y=0} \\ &= \frac{\Delta}{2\pi}\ \cos^{-1}\frac{x}{a} \end{aligned} \tag{G38}$$

Eqs. (G37) and (G38) agree with the known solution **(Barenblatt, 1962)** (see **page 3.11** and the dashed curve in **Fig. G24**).

In the preceding example, we discussed the case where the position of the point of separation of the crack surface and the wedge is definite and known beforehand. For the case where the position of the separating point is not known, we can still apply the present method by simply imposing an additional condition at the point of separation. Some discussion on a problem of this type is found in Barenblatt's review (**Barenblatt, 1962**). The following simple examples illustrate the applicability of the method.

Consider a semi-infinite crack whose elastic field is characterized by the stress intensity factor K_{appl}. When a thin, smooth, rigid wedge with uniform thickness, Δ, is inserted into the crack up to the leading edge, the contact between the crack surface and the wedge will occur over a portion, ℓ, near the crack tip, as shown in **Fig. G17**, and the surfaces will separate outside this portion since the crack opening profile is parabolic before the insertion of the wedge.

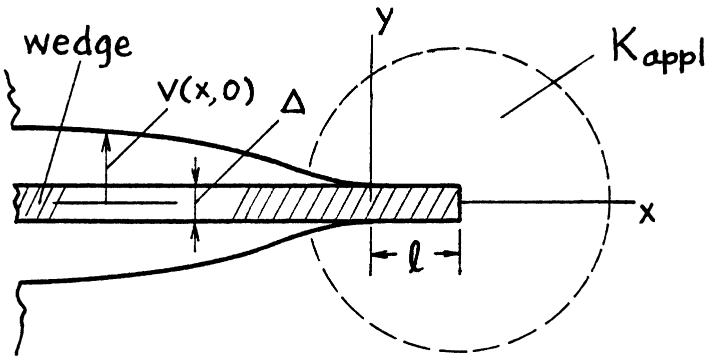


Fig. G17

To apply the present method to this example, we interpret the problem in the following way. First, consider a semi-infinite crack filled with the wedge. We apply a remote external load that would generate the stress intensity factor K_{appl} if the wedge is not inserted. The separation of the contact surfaces will occur at a certain distance, ℓ, from the tip. For convenience, the coordinates are taken as shown in **Fig. G17**. $Z_D(z)$ and $\sigma_{yD}(x,0)$ for this case are

$$Z_D(z) = \frac{E'\Delta}{4\pi}\frac{1}{z-\ell} \tag{G39}$$

and

$$\sigma_{yD}(x,0) = \frac{E'\Delta}{4\pi}\frac{1}{x-\ell} \tag{G40}$$

$Z_G(z,x')$ for a semi-infinite crack is known as (see **page 3.6**)

$$Z_G(z,x') = \frac{1}{\pi}\sqrt{\frac{-x'}{z}}\cdot\frac{1}{z-x'} \tag{G41}$$

Thus $Z_C(z)$ is calculated as

$$\begin{aligned} Z_C(z) &= \frac{E'\Delta}{4\pi^2}\frac{1}{\sqrt{z}}\int_{-\infty}^{0}\frac{1}{x'-\ell}\cdot\frac{\sqrt{-x'}}{z-x'}\,dx' = \frac{E'\Delta}{4\pi^2}\frac{1}{\sqrt{z}}\int_{0}^{\infty}\frac{1}{\ell+x'}\cdot\frac{\sqrt{x'}}{z+x'}\,dx' \\ &= -\frac{E'\Delta}{4\pi}\left[\sqrt{\frac{\ell}{z}}\frac{1}{\ell-z}-\frac{1}{\ell-z}\right] \end{aligned} \tag{G42}$$

Note that the function

$$Z_D(z)+Z_C(z) = \frac{E'\Delta}{4\pi}\frac{1}{z-\ell}\sqrt{\frac{\ell}{z}} \tag{G43}$$

is the solution to the problem where the portion of the wedge that is located $x<0$ is removed from the crack (**Fig. G18**). This solution is also obtained using the finite wedge which fills a single crack, as follows (**Fig. G19**). For this case (see **Eq. (G6a)**)

$$Z_D(z) = \frac{E'\Delta}{4\pi}\left(\frac{1}{z-\ell}-\frac{1}{z}\right) \tag{G44}$$

$$\sigma_{yD}(x,0) = \frac{E'\Delta}{4\pi}\left(\frac{1}{x-\ell}-\frac{1}{x}\right) \tag{G45}$$

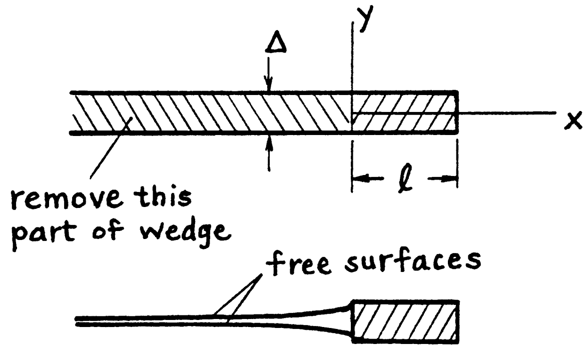


Fig. G18

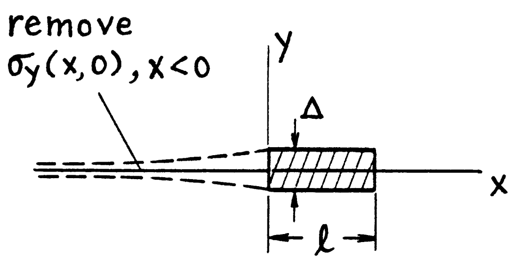


Fig. G19

$Z_G(z,x')$ is the same as **Eq. (G41)**. Thus

$$\begin{aligned} Z_C(z) &= \int_{-\infty}^{0} \sigma_{yD}\left(x',0\right) Z_G(z,x')dx \\ &= \frac{E'\Delta}{4\pi}\frac{1}{z-\ell}\sqrt{\frac{\ell}{z}} - Z_D(z) \end{aligned}$$

$$Z_D(z) + Z_C(z) = \frac{E'\Delta}{4\pi}\frac{1}{z-\ell}\sqrt{\frac{\ell}{z}} \qquad \text{(G46)}$$

Eq. (G46) agrees with **Eq. (G43)**.

Since we have an additional crack-stress field, $Z_{C\,appl}(z)$, due to applied external load, which is given by (see **page 3.1**)

$$Z_{C\,appl}(z) = \frac{K_{appl}}{\sqrt{2\pi z}} \qquad \text{(G47)}$$

the total stress function, $Z(z)$, is given by

$$\begin{aligned} Z(z) &= Z_D(z) + Z_C(z) + Z_{C\,appl}(z) \\ &= \frac{E'\Delta}{4\pi}\frac{1}{z-\ell}\sqrt{\frac{\ell}{z}} + \frac{K_{appl}}{\sqrt{2\pi z}} \end{aligned} \tag{G48}$$

The unknown length of contact, ℓ, is obtained in terms of Δ and K_{appl} from the boundary condition at the point of separation. The condition is that the separation of surfaces occurs smoothly, or that the stress singularity at the separation point, $K_{x=0}$, is zero. That is, from **Eq. (G48)**,

$$-\frac{E'\Delta}{2\sqrt{2\pi\ell}} + K_{appl} = 0$$

or

$$\ell = \frac{1}{8\pi}\left(\frac{E'\Delta}{K_{appl}}\right)^2 \tag{G49}$$

Eq. (G48) is then simplified as

$$\begin{aligned} Z(z) &= \frac{E'\Delta}{4\pi}\frac{1}{z-\ell}\sqrt{\frac{z}{\ell}} \\ \text{or} \quad &= \frac{K_{appl}}{\sqrt{2\pi}}\frac{\sqrt{z}}{z-\ell} \end{aligned} \tag{G50}$$

The function $\overline{Z}(z) = \int Z(z)dz$ is also written in a simple form as

$$\overline{Z}(z) = \frac{E'\Delta}{2\pi}\left\{\sqrt{\frac{z}{\ell}} + \frac{1}{2}\ell\text{n}\left(\frac{\sqrt{z/\ell}-1}{\sqrt{z/\ell}+1}\right)\right\} \tag{G51}$$

The stress distribution, $\sigma_y(x,0)$, and the displacement, $v(x,0)$, along the x-axis are calculated from **Eqs. (G50) and (G51)**, respectively, as

$$\sigma_y(x,0) = \ \text{Re}\{Z(z)\}_{y=0} = \begin{cases} 0 & x < 0 \\ \dfrac{E'\Delta}{4\pi\sqrt{\ell}}\dfrac{\sqrt{x}}{x-\ell}\left(= \dfrac{K_{appl}}{\sqrt{2\pi}}\dfrac{\sqrt{x}}{x-\ell}\right) & x \geq 0 \end{cases} \tag{G52}$$

$$v(x,0) = \frac{2}{E'}\ \text{Im}\{\overline{Z}(z)\}_{y=0} = \begin{cases} \frac{\Delta}{\pi}\left(\sqrt{-\frac{x}{\ell}} + \cot^{-1}\sqrt{-\frac{x}{\ell}}\right) & x \leq 0 \\ \Delta/2 & 0 \leq x \leq \ell \\ 0 & x > \ell \end{cases} \tag{G53}$$

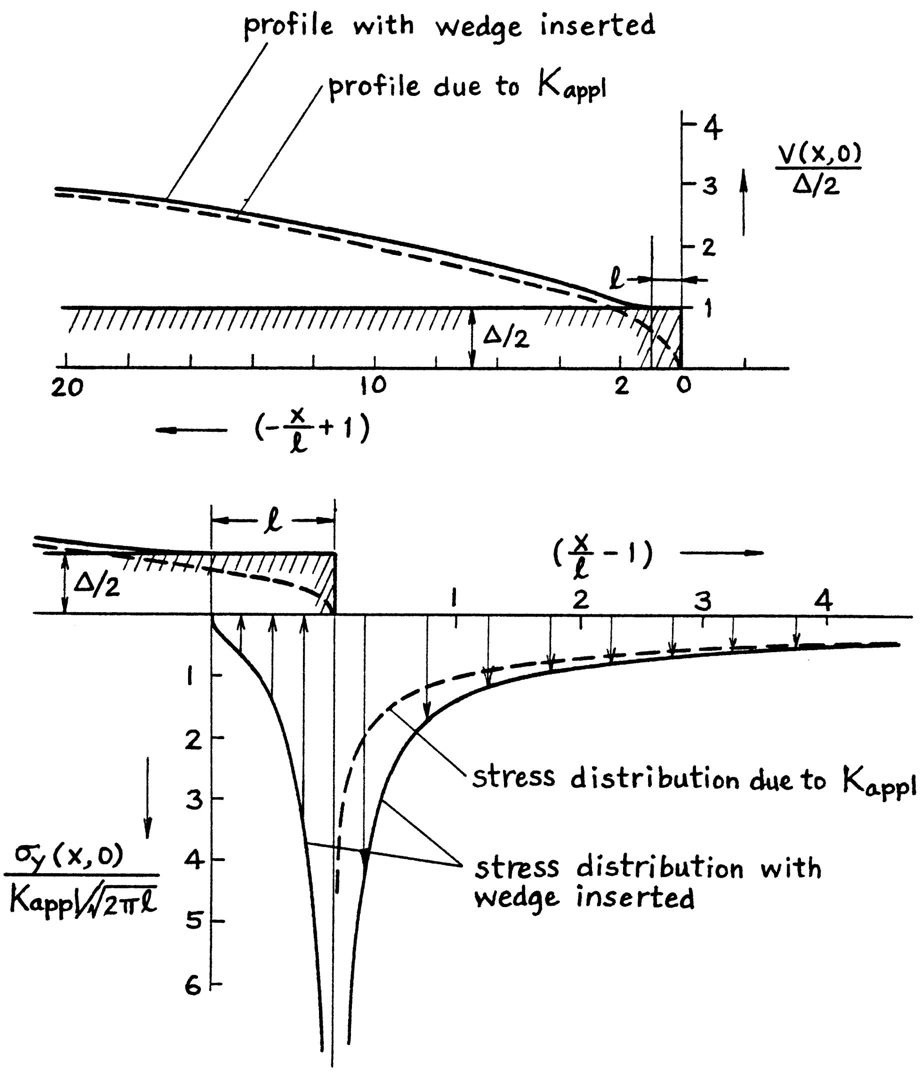


Fig. G20. Crack profile and stress distribution given by Eqs. (G53) and (G52), respectively.

The stress distribution and displacement given by **Eqs. (G52) and (G53)** are shown in **Fig. G20**, where the stress distribution and crack profile due to the applied force (K_{appl}) are also shown by dashed lines for comparison.

It is noted from **Eq. (G49)** that the contact between the wedge and the crack surfaces always exists ($\ell > 0$), as expected, regardless of the magnitude of applied load.

As another simple example of this type, we again consider the situation given by **Fig. G16**. In addition to the insertion of the wedge, the crack is now assumed to be subjected to a remote uniform compressive stress, p, as shown in **Fig. G21**.

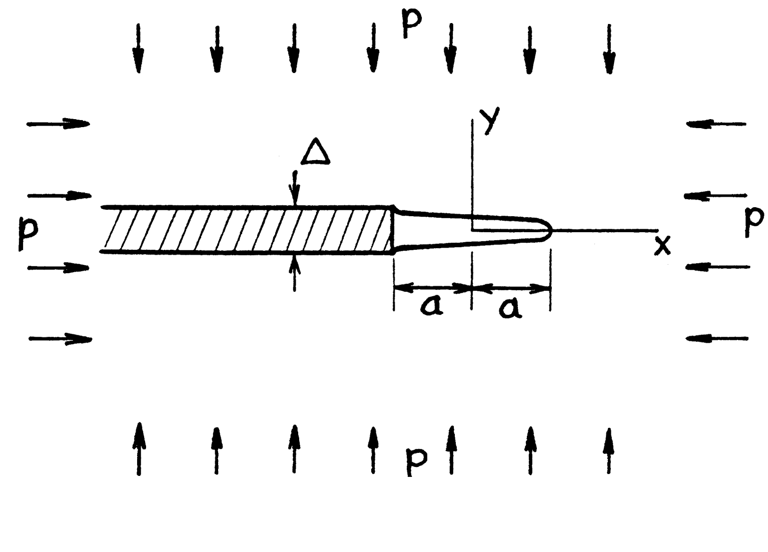


Fig. G21

The stress function is readily obtained by the superposition of **Eq. (G36)** and the well-known solution for the Griffith crack (**page 5.1**). Thus the total stress functions are written as

$$Z(z) = \frac{E'\Delta}{4\pi}\frac{1}{\sqrt{z^2 - a^2}} - \frac{p}{\sqrt{1 - (a/z)^2}} \tag{G54a}$$

$$\overline{Z}(z) = \frac{E'\Delta}{4\pi}\cosh^{-1}\frac{z}{a} - p\sqrt{z^2 - a^2} \tag{G54b}$$

The stress intensity factors at $x = \pm a$ are

$$K_{+a} = \frac{E'\Delta}{4\sqrt{\pi a}} - p\sqrt{\pi a} \tag{G55a}$$

$$K_{-a} = -\frac{E'\Delta}{4\sqrt{\pi a}} - p\sqrt{\pi a} \tag{G55b}$$

While $K_{-a} < 0$ always holds, from **Eq. (G55a)**,

$$K_{+a} \leq 0 \quad \text{when } p \geq p_0 = \frac{E'\Delta}{4\pi a} \tag{G56}$$

Therefore, **Eqs. (G54) and (G55)** are valid as they are when $p \leq p_0 = E'\Delta/(4\pi a)$. When the compressive stress p exceeds p_0, the crack closure will occur over a certain length, ℓ, from the right end of the crack, as shown in **Fig. G22b**. The length, ℓ, and the resulting elastic field are easily obtained by imposing the condition for smooth contact between the closed surfaces.

a. $p = p_0 = \dfrac{E'\Delta}{4\pi a}$

b. $p > p_0$

$a' = \dfrac{E'\Delta}{4\pi p}$

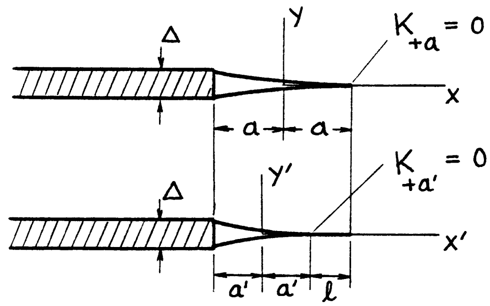


Fig. G22

We first examine the situation shown in **Fig. G22a**, where $p = p_0$. We can immediately write stress functions and $K_{\pm a}$ from **Eqs. (G54) and (G55)**.

$$Z(z) = -p_0\sqrt{\frac{z-a}{z+a}} \tag{G57a}$$

$$\overline{Z}(z) = p_0 a\left(\cosh^{-1}\frac{z}{a} - \sqrt{\left(\frac{z}{a}\right)^2 - 1}\right) \tag{G57b}$$

$$K_{+a} = 0 \tag{G58a}$$

$$K_{-a} = -2p_0\sqrt{\pi a} = -\frac{E'\Delta}{2\sqrt{\pi a}}\left(\text{or} = -\sqrt{E'\Delta p_0}\right) \tag{G58b}$$

Now the complete solution for $p > p_0$ (**Fig. G22b**) can be expressed in the same form simply replacing z, a and p_0 by z', a' and p, respectively, where $z' = z + \ell/2$, $a' = E'\Delta/(4\pi p)$, and $\ell = 2(a - a')$. Thus

$$Z(z') = -p\sqrt{\frac{z'-a'}{z'+a'}} \tag{G59a}$$

$$\overline{Z}(z') = \frac{E'\Delta}{4\pi}\left(\cosh^{-1}\frac{z'}{a'} - \sqrt{\left(\frac{z'}{a'}\right)^2 - 1}\right) \tag{G59b}$$

$$K_{+a'} = 0 \tag{G60a}$$

$$K_{-a'} = -\sqrt{E'p\Delta}\left(= -2p\sqrt{\pi a'} = -\frac{E'\Delta}{2\sqrt{\pi a'}}\right) \tag{G60b}$$

where

$$p = \frac{E'\Delta}{4\pi a'}$$

The length of closure, ℓ, is given by

$$\ell = 2(a - a') = 2\left(a - \frac{E'\Delta}{4\pi p}\right) = \frac{E'\Delta}{4\pi}\left(\frac{1}{p_0} - \frac{1}{p}\right) \tag{G61}$$

When an internal pressure, p, is applied inside the crack instead of a remote compression, it is readily shown that the contact between the surfaces along $x < -a$ is lost completely when the internal pressure p exceeds $p_0 = E'\Delta/(4\pi a)$.

CRACKS COLLINEARLY LOCATED WITH DISLOCATIONS

(Combinations of Displacement-Specified Cracks and Stress-Specified Cracks)

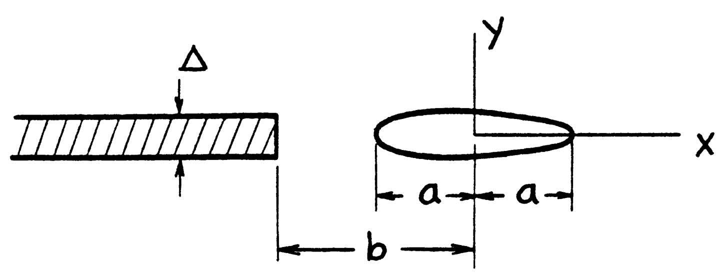


Fig. G23

In the very simple example shown in **Fig. G23**, a uniform dislocation with thickness Δ is located along $x \le -b,\ y = 0$ and a single crack is located at $|x| \le a,\ y = 0$. This situation is realized when a semi-infinite crack located at $x \le -b,\ y = 0$ is filled with a thin, smooth, rigid wedge with uniform thickness Δ, leaving another crack stress free.

$Z_D(z)$ for this case is given, again from **Eq. (G4a)**, by

$$Z_D(z) = \frac{E'\Delta}{4\pi}\frac{1}{z+b} \tag{G62}$$

Then $\sigma_{yD}(x,0)$ is readily given by

$$\sigma_{yD}(x,0) = \frac{E'\Delta}{4\pi}\frac{1}{x+b} \tag{G63}$$

Since the Green's function, $Z_G(z,x')$, given by **Eq. (G34)** is again valid,

$$Z_C(z) = \int_C \sigma_{yD}(x',0)Z_G(z,x')dx' = \frac{E'\Delta}{4\pi^2}\frac{1}{\sqrt{z^2-a^2}}\int_{-a}^{a}\frac{\sqrt{a^2-x'^2}}{(z-x')(x'-b)}dx'$$

Performing the integration, we can obtain the following final expression for $Z_C(z)$:

$$Z_C(z) = \frac{E'\Delta}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\left(1-\frac{\sqrt{b^2-a^2}}{z+b}\right) - \frac{E'\Delta}{4\pi}\frac{1}{z+b} \tag{G64}$$

Thus, the total stress function is

$$Z(z) = Z_D(z) + Z_C(z) = \frac{E'\Delta}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\left(1-\frac{\sqrt{b^2-a^2}}{z+b}\right) \tag{G65a}$$

and

$$\overline{Z}(z) = \int Z(z)dz = \frac{E'\Delta}{4\pi}\left\{\cosh^{-1}\frac{z}{a} - i\cos^{-1}\frac{a^2+bz}{a(b+z)}\right\} \tag{G65b}$$

The stress intensity factors at $x = \pm a$ are

$$K_{+a} = \frac{E'\Delta}{4\sqrt{\pi a}}\left(1-\sqrt{\frac{b-a}{b+a}}\right) \tag{G66a}$$

and

$$K_{-a} = -\frac{E'\Delta}{4\sqrt{\pi a}}\left(1-\sqrt{\frac{b+a}{b-a}}\right) \tag{G66b}$$

When $a = b$, **Eq. (G65)** is reduced to **Eq. (G36)**, that is, the solution for the situation of **Fig. G16**. Therefore, it is seems interesting to examine how the profile of the crack changes when the ratio a/b varies. The profile is calculated from **Eq. (G65b)** using **Eq. (G3)**.

$$v(x,0) = \frac{2}{E'}\ \mathrm{Im}\{\overline{Z}(z)\}_{y=0} = \frac{\Delta}{2\pi}\left\{\cos^{-1}\frac{x}{a} - \cos^{-1}\frac{a^2+bx}{a(b+x)}\right\} \tag{G67}$$

The shape of the crack given by **Eq. (G67)** is shown in **Fig. G24** for various values of a/b while the distance b is fixed for convenience.

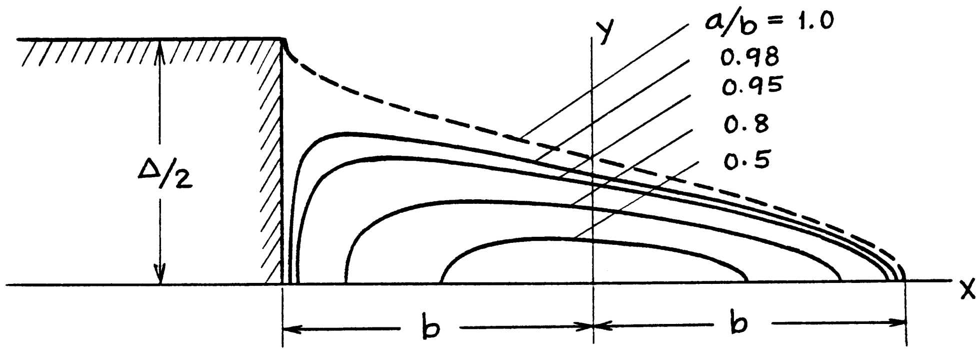


Fig. G24

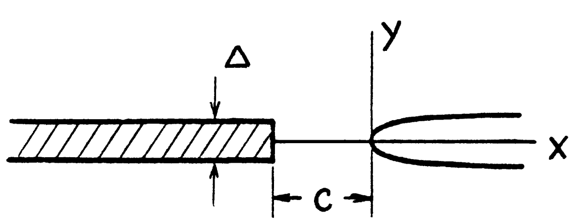
When a is very small compared with b, the elastic field near the crack should correspond to a Griffith crack under uniform tension $\sigma_y(0,0) = E'\Delta/(4\pi b)$. In fact, when $a/b \ll 1$ and $z/b \ll 1$, **Eq. (G65a)** is reduced to the following well-known solution:

$$Z(z) = \frac{E'\Delta}{4\pi b}\frac{1}{\sqrt{1-\left(\frac{a}{z}\right)^2}} = \frac{\sigma_y(0,0)}{\sqrt{1-\left(\frac{a}{z}\right)^2}} \tag{G68}$$

It should be noted that the solution, **Eq. (G65)**, is very useful in constructing solutions for various problems associated with a single stress-free (or stress-specified) crack and dislocations with uniform thickness. It is possible to derive solutions by superposition of **Eq. (G65)** and the limiting procedure without performing the integration of **Eq. (G30)**. Westergaard functions, $Z(z)$, for a few simple examples, which are directly obtained from **Eq. (G65)**, are illustrated next.

EXAMPLES OF WESTERGAARD FUNCTIONS ASSOCIATED WITH UNIFORM DISLOCATIONS AND A STRESS-FREE CRACK

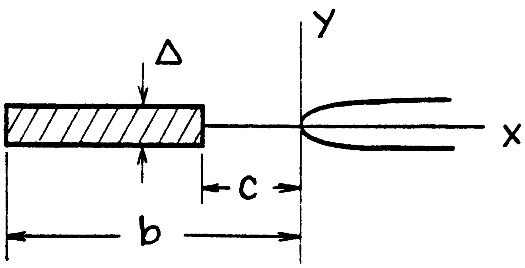
a. Two opposing cracks, one of which is filled with an infinite uniform wedge, **Fig. G25(a)**.



$$Z(z) = \frac{E'\Delta}{4\pi}\frac{\sqrt{c}}{(z+c)\sqrt{-z}} \tag{G69}$$

Fig. G25(a)

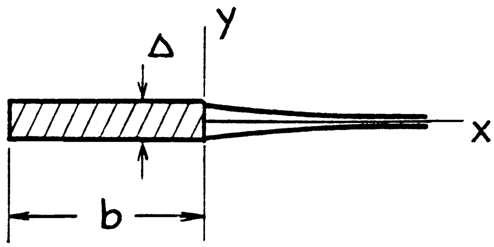
b. A finite crack filled with uniform wedge and a semi-infinite crack, **Fig. G25(b)**.



$$Z(z) = \frac{E'\Delta}{4\pi}\frac{1}{\sqrt{-z}}\left(\frac{\sqrt{c}}{z+c} - \frac{\sqrt{b}}{z+b}\right) \qquad (\mathbf{G70})$$

Fig. G25(b)

c. A semi-infinite crack, the tip of which is filled with a uniform wedge with finite length, **Fig. G25(c)**.

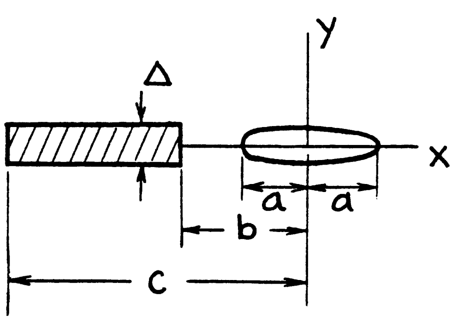


$$Z(z) = -\frac{E'\Delta}{4\pi}\frac{\sqrt{b}}{(z+b)\sqrt{-z}} \qquad (\mathbf{G71})$$

See **Eq. (G43) or (G46)**.

Fig. G25(c)

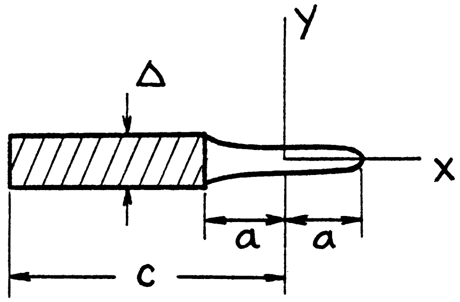
d. Two finite cracks, one of which is filled with a uniform wedge, **Fig. G25(d)**.



$$Z(z) = \frac{E'\Delta}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\left(\frac{\sqrt{c^2-a^2}}{z+c} - \frac{\sqrt{b^2-a^2}}{z+b}\right) \qquad (\mathbf{G72})$$

Fig. G25(d)

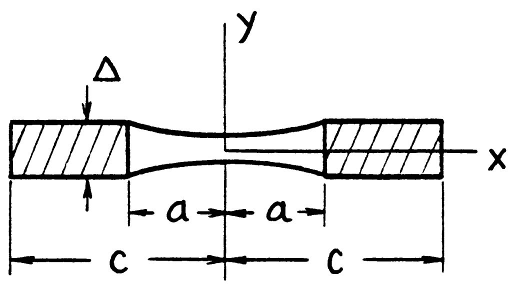
e. A finite crack partially filled with a uniform wedge at one end, **Fig. G25(e)**.



$$Z(z) = \frac{E'\Delta}{4\pi}\frac{1}{z+c}\sqrt{\frac{c^2-a^2}{z^2-a^2}} \qquad \text{(G73)}$$

Fig. G25(e)

f. A finite crack partially filled with two identical uniform wedges at both ends, **Fig. G25(f)**.



$$Z(z) = \frac{E'\Delta}{4\pi}\frac{z}{z^2-c^2}\frac{\sqrt{c^2-a^2}}{\sqrt{z^2-a^2}} \qquad \text{(G74)}$$

Fig. G25(f)

g. A finite crack filled with a uniform wedge subjected to remote tension, **Fig. G25(g)**.

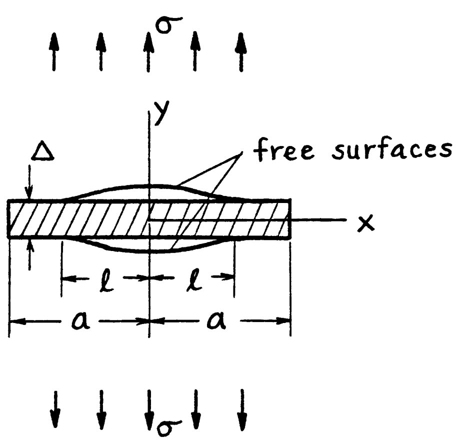


Fig. G25(g)

When $\sigma \le \frac{E'\Delta}{2\pi a}$,

$$\left.\begin{aligned} \ell &= 0 \\ Z(z) &= \frac{E'\Delta}{2\pi}\frac{a}{z^2-a^2} + \sigma \end{aligned}\right\} \qquad \text{(G75)}$$

See **Eq. (G6a)**.

When $\sigma > \frac{E'\Delta}{2\pi a}$

$$\left.\begin{aligned} \ell &= a\sqrt{1-\left(\frac{E'\Delta}{2\pi\sigma a}\right)^2} \\ Z(z) &= \sigma\frac{z}{z^2-a^2}\sqrt{z^2-\ell^2} \end{aligned}\right\} \qquad \text{(G76)}$$

where 2ℓ is the length of separated (free) surfaces.

THE GREEN'S FUNCTION FOR A SINGLE STRESS-FREE CRACK AND DISLOCATIONS WITH ARBITRARY SHAPE

Solutions are easily obtained from **Eq. (G65)** by superposition for many other problems in which there is only one stress-free crack or only one stress-free portion along the x-axis, such as shown in **Fig. G25, (c), (e), and (f)**. This suggests the use of **Eq. (G65)** as the Green's function for such problems. When the same portion is subjected to a specified stress instead of stress free, the ordinary method of crack stress analysis is separately applied. Then the result must be superimposed on the solution obtained under the stress-free condition. Note that the cases illustrated by **Fig. G25, (c), (e), and (f)** are simply treated as special cases of the single stress-free crack problem. Therefore, we address only the Green's function for the problem associated with a single stress-free crack and dislocations (or cracks) with arbitrarily prescribed shape.

For convenience, replacing b by $-x'$ we rewrite **Eq. (G65a)** as

$$Z_0(z,x') = \frac{E'\Delta}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\left(1-\frac{\sqrt{x'^2-a^2}}{z-x'}\right) \tag{G77}$$

Recalling the previous discussion leading to **Eq. (G16)**, the Green's function is readily derived as

$$\begin{aligned} Z_G(z,x') &= \lim_{\substack{\varepsilon\to 0\\ \Delta\varepsilon = d}}\left\{Z_0(z,x') - Z_0(z,x'-\varepsilon)\right\} \\ &= \frac{d}{\Delta}\frac{d}{dx'}\left\{Z_0(z,x')\right\} \end{aligned}$$

that is,

$$Z_G(z,x') = -\frac{E'd}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\frac{d}{dx'}\left(\frac{\sqrt{x'^2-a^2}}{z-x'}\right) \tag{G78}$$

or

$$\begin{aligned} Z_G(z,x') &= -\frac{E'd}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\left\{\frac{x'}{\sqrt{x'^2-a^2}}\frac{1}{z-x'}+\frac{\sqrt{x'^2-a^2}}{(z-x')^2}\right\} \\ &= -\frac{E'd}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\cdot\frac{zx'-a^2}{(z-x')^2\sqrt{x'^2-a^2}} \end{aligned} \tag{G79}$$

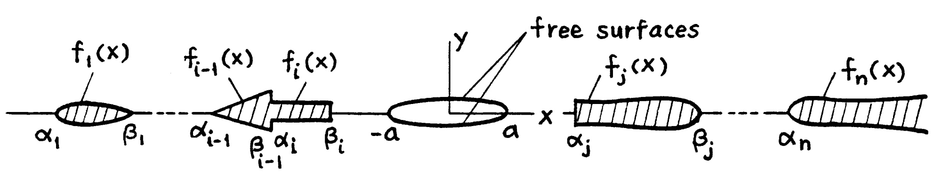


Fig. G26

When symmetric vertical displacement, $v(x,0)$, is specified along the x-axis, except in the region $|x| \leq a$ where a stress-free condition is given (**Fig. G26**), $Z(z)$ is obtained by the following integration. Note that in the region $|x| \leq a$ we can simply assume $v(x,0) = 0$.

$$Z(z) = \frac{1}{d}\int_{-\infty}^{\infty} v(x',0) Z_G(z,x')dx' \tag{G80}$$

or using the same notations used in **Fig. G7** and **Eq. (G19)**

$$Z(z) = \frac{1}{d}\sum_{i=1}^{n}\int_{\alpha_i}^{\beta_i} f_i(x') Z_G(z,x')dx' \tag{G81}$$

When **Eq. (G79)** is used for $Z_G(z,x')$, **Eq. (G80)** becomes

$$Z(z) = -\frac{E'}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\int_{-\infty}^{\infty} v(x',0)\cdot\frac{zx'-a^2}{(z-x')^2\sqrt{x'^2-a^2}}dx' \tag{G82}$$

and when **Eq. (G78)** is used, **Eq. (G80)** is integrated by parts,

$$\begin{aligned} Z(z) &= -\frac{E'}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\int_{-\infty}^{\infty} v(x',0)\frac{d}{dx'}\left(\frac{\sqrt{x'^2-a^2}}{z-x'}\right)dx' \\ &= -\frac{E'}{4\pi}\frac{1}{\sqrt{z^2-a^2}}\left[v(x',0)\frac{\sqrt{x'^2-a^2}}{z-x'}\Bigg|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} v'(x',0)\frac{\sqrt{x'^2-a^2}}{z-x'}dx'\right] \end{aligned} \tag{G83}$$

where $v'(x',0) = \frac{d}{dx'}v(x',0)$. The choice between **Eqs. (G82) and (G83)** can be made for simplicity of integration. Note that this method can be used even when $\beta_i = -a$ and/or $\alpha_j = a$ in **Fig. G26**.

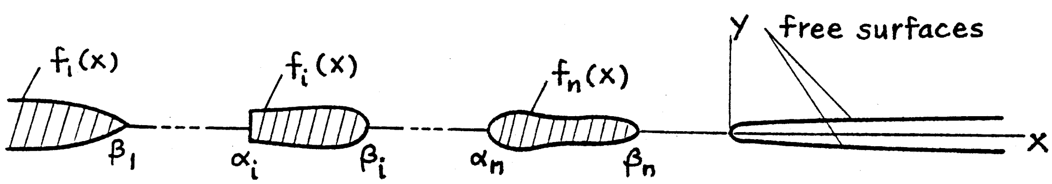


Fig. G27

When this stress-free crack is a semi-infinite crack located along the positive x-axis (**Fig. G27**), in a similar manner, starting with **Eq. (G69)**, or as a limiting case of the above, the Green's function is obtained as follows. $Z_0(z,x')$ in this case is

$$Z_0(z,x') = \frac{E'\Delta}{4\pi}\frac{\sqrt{-x'}}{(z-x')\sqrt{-z}} \tag{G84}$$

The Green's function is

$$\begin{aligned} Z_G(z,x') &= \frac{d}{\Delta}\frac{d}{dx'}\{Z_0(z,x')\} \\ &= \frac{E'd}{4\pi}\frac{1}{\sqrt{-z}}\frac{1}{dx'}\left(\frac{\sqrt{-x'}}{z-x'}\right) \end{aligned} \quad (\mathbf{G85})$$

or

$$\begin{aligned} &= -\frac{E'd}{4\pi}\frac{1}{\sqrt{-z}}\left\{\frac{1}{2}\frac{1}{\sqrt{-x'}(z-x')}-\frac{\sqrt{-x'}}{(z-x')^2}\right\} \\ &= -\frac{E'd}{4\pi}\frac{1}{\sqrt{-z}}\cdot\frac{z+x'}{(z-x')^2\sqrt{-z}} \end{aligned} \quad (\mathbf{G86})$$

Then, $Z(z)$ for the situation shown in **Fig. G27**, again using the same notation, is given by the integral

$$Z(z) = \frac{1}{d}\int_{-\infty}^{0} v(x',0)Z_G(z,x')dx' \quad (\mathbf{G87})$$

or

$$Z(z) = \frac{1}{d}\sum_{i=1}^{n}\int_{\alpha_i}^{\beta_i} f_i(x')Z_G(z,x')dx' \quad (\mathbf{G88})$$

Using **Eq. (G86)**

$$Z(z) = -\frac{E'}{4\pi}\frac{1}{\sqrt{-z}}\int_{-\infty}^{0} v(x',0)\cdot\frac{z+x'}{(z-x')^2\sqrt{-x'}}dx' \quad (\mathbf{G89})$$

or using **Eq. (G85)**,

$$\begin{aligned} Z(z) &= \frac{E'}{4\pi}\frac{1}{\sqrt{-z}}\int_{-\infty}^{0} v(x',0)\frac{d}{dx'}\left(\frac{\sqrt{-x'}}{z-x'}\right)dx' \\ &= \frac{E'}{4\pi}\frac{1}{\sqrt{-z}}\left[v(x',0)\frac{\sqrt{-x'}}{z-x'}\bigg|_{-\infty}^{0} - \int_{-\infty}^{0} v'(x',0)\frac{\sqrt{-x'}}{z-x'}dx'\right] \end{aligned} \quad (\mathbf{G90})$$

where $v'(x',0) = \frac{d}{dx'}v(x',0)$.

The choice between **Eqs. (G89) and (G90)** is again based on simplicity of integration. This method applies to a special case of $\beta_n = 0$ in **Fig. G27**.

The method discussed in this section combines the two integrals, **Eq. (G18) or (G19) and Eq. (G30)**, into a single integral. **Equation (G18) or (G19)** is needed to calculate $\sigma_y(x,0)$, which is then used in **Eq. (G30)**. Thus the general method given by **Eq. (G30)** was further simplified for problems involving a single stress-free crack and cracks or dislocations with arbitrary shape.

The method is easily extended to problems where there are more than one stress-specified cracks, along with cracks or dislocations with arbitrarily specified shape.

SUMMARY

This appendix presents a simple method for obtaining the Westergaard stress function for dislocations and cracks.

First, problems associated with dislocations with uniform height (or cracks with uniform displacement) were discussed. Then, to extend the method's application to dislocations (or cracks) with arbitrarily prescribed shape, the Green's function was obtained. Thus no more than integrations were required to obtain the Westergaard stress function. Examples of $Z(z)$ and $\overline{Z}(z)$ for various dislocations were given.

The method was then further extended to problems with a combination of displacement-specified cracks (or dislocations) and stress-specified cracks. The general approach was presented and its applications to a few problems were discussed in detail. Many examples of $Z(z)$ were compiled.

Finally, the Green's functions for problems involving both displacement-specified and stress-specified cracks were established for a few simple configurations, further simplifying the general method.

The method presented in this appendix requires only integrations, and thus provides a simple way of treating problems associated with collinear dislocations and cracks.

APPENDIX H

THE PLASTIC ZONE INSTABILITY CONCEPT APPLIED TO ANALYSIS OF PRESSURE VESSEL FAILURE[1]

This appendix reviews the plastic zone instability concept and refines the procedures for determining the conditions for instability. The plastic zone instability failure criterion is applied to several crack configurations in a specific thin-walled cylindrical pressure vessel with semi-spherical end caps. The cracks analyzed are a longitudinal and a circumferential through-wall cracks in a cylindrical shell, and a through-wall crack in a spherical shell. The results predict the possibility that for ductile materials in certain combinations of geometry and loading the failure controlled by the plastic zone instability may precede the failure controlled by the plastic collapse load. Perhaps the results suggest the necessity, in the presence of cracks, for consideration of the plastic zone instability along with the failure criterion based on the plastic limit load.

INTRODUCTION

In the evaluation of mechanical integrity of pressure vessels, the stability analysis of through-wall cracks is documented in various international sources [2]. In these sources, the dual approach for the following two conditions is generally regarded as reasonable.

1. Avoidance of conditions for unstable running crack growth
2. Avoidance of conditions for unrestricted plastic flow (or for plastic collapse)

The plastic zone instability analysis discussed in this appendix addresses the second condition.

The concept of the plastic zone instability (PZI) was proposed by Vazquez in 1971 **(Vazquez, 1971)**, but it has not received much attention. It seems that the only extensive application of this method is found in the work of Tada and Paris **(Tada, 1983c)**. Some of the results obtained in **Tada (1983c)** are included in this discussion.

The PZI concept is based on the plastic zone size (r_Y) corrected linear-elastic fracture mechanics (LEFM) analysis, and therefore only LEFM solutions are required. In what follows, the LEFM solutions are presented first for the crack configurations of interest for convenience of subsequent discussions. Then the r_Y-correction

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[1] **Tada 1996.**

[2] **For example: British Standards Institute PD 64-93-91; U.K. Central Electricity Generating Board R-6; International Insitute of Welding 11S-SST 1157-90; American Petroleum Institute RP579; and Materials Properites Council (USA), Fitness for Service Report.**